

Weather Forecasting

Presented by: Mehak, Tanya, Kalyani, Rishitha (Group 9)

📍 Introduction



Weather forecasting is a crucial aspect of daily life, impacting various sectors such as agriculture, transportation, and disaster management. Traditional forecasting methods rely heavily on numerical simulations, which can be complex and computationally expensive.

This project aims to leverage machine learning techniques to classify weather conditions (Sunny, Cloudy, Rainy, Snowy) based on meteorological features such as temperature, humidity, wind speed, precipitation, and atmospheric pressure. By evaluating multiple classification models, we identify the most accurate approach to automate weather classification, making predictions faster and more efficient.

📍 Real-World Beneficiaries

- Agriculture & Farming: Enhanced crop management, reduced risk of crop loss.
- Disaster Management & Emergency Services: Better preparedness for extreme weather, reduced casualties, and infrastructure damage.
- Transportation & Logistics: Minimized disruptions, safer travel, fewer delays.
- Energy Sector: Optimized renewable energy management (solar, wind) for better grid stability.
- Public Safety & Health: Timely public alerts reducing health risks associated with severe weather events.



Enhancing Weather Forecasting through ML

How This Project Improves the Current Situation



- **Improved Accuracy:** Leveraging ensemble ML models (XGBoost, LightGBM, CatBoost) surpasses traditional statistical methods in prediction quality.
- **Reliable Predictions:** Advanced feature engineering ensures predictions are based on key meteorological insights.
- **Balanced & Fair Results:** Application of SMOTE addresses data imbalance, improving model fairness, particularly for rare weather conditions.
- **Scalable & Efficient:** ML-driven weather classification can be seamlessly integrated into automated forecasting systems, efficiently analyzing large datasets.

📍 Dataset Overview

Dataset Size & Structure:

- Total Samples: 13,200 entries
- Total Features: 11 (7 numerical, 4 categorical)
- Weather Classes: 4 (Sunny, Cloudy, Rainy, Snowy)

```
RangeIndex: 13200 entries, 0 to 13199
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Temperature                           13200 non-null  float64
1   Humidity                              13200 non-null  int64
2   Wind Speed                            13200 non-null  float64
3   Precipitation (%)                     13200 non-null  float64
4   Cloud Cover                           13200 non-null  object
5   Atmospheric Pressure                  13200 non-null  float64
6   UV Index                             13200 non-null  int64
7   Season                                13200 non-null  object
8   Visibility (km)                       13200 non-null  float64
9   Location                              13200 non-null  object
10  Weather Type                           13200 non-null  object
dtypes: float64(5), int64(2), object(4)
```

Numerical Features



- Temperature
- Humidity
- Wind Speed
- Precipitation (%)
- Atmospheric Pressure
- UV Index
- Visibility (km)

Categorical Features

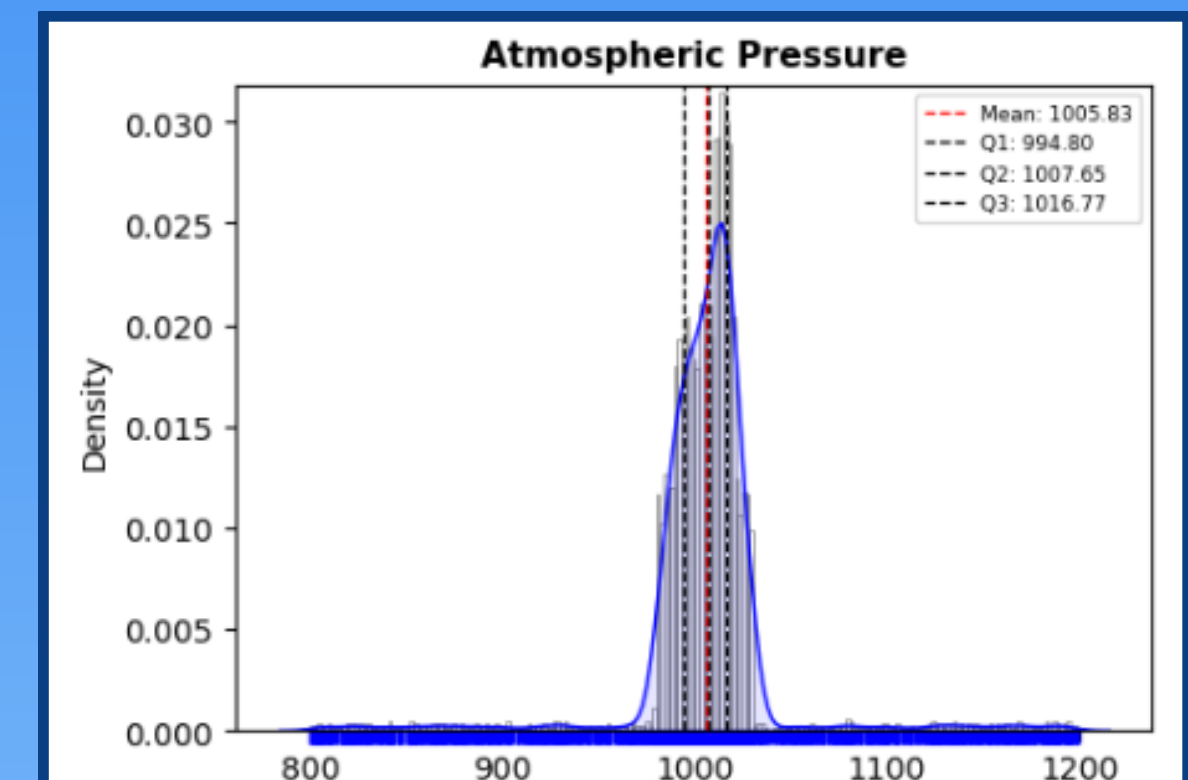
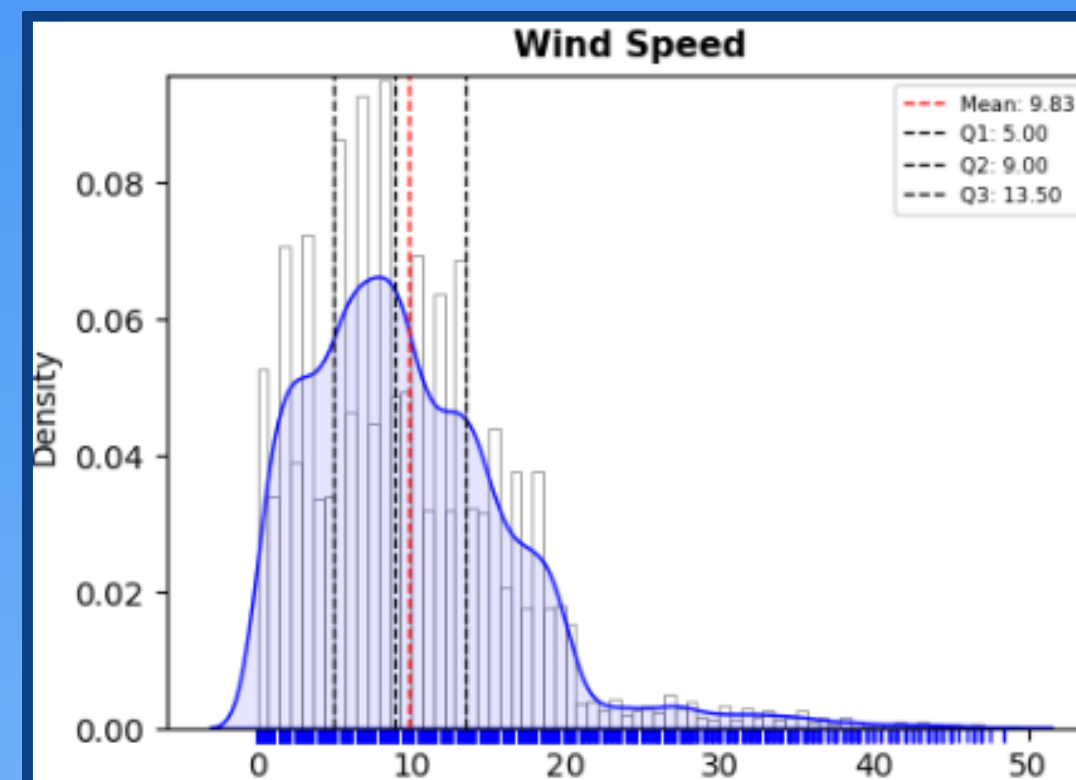
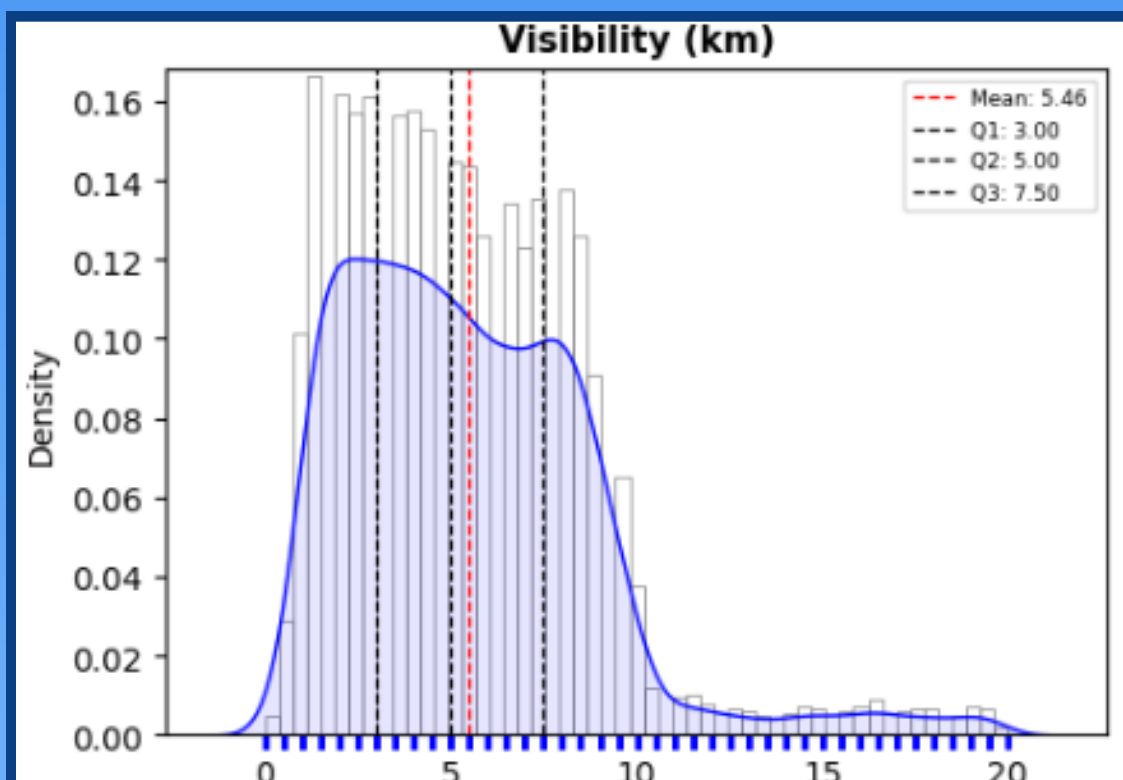
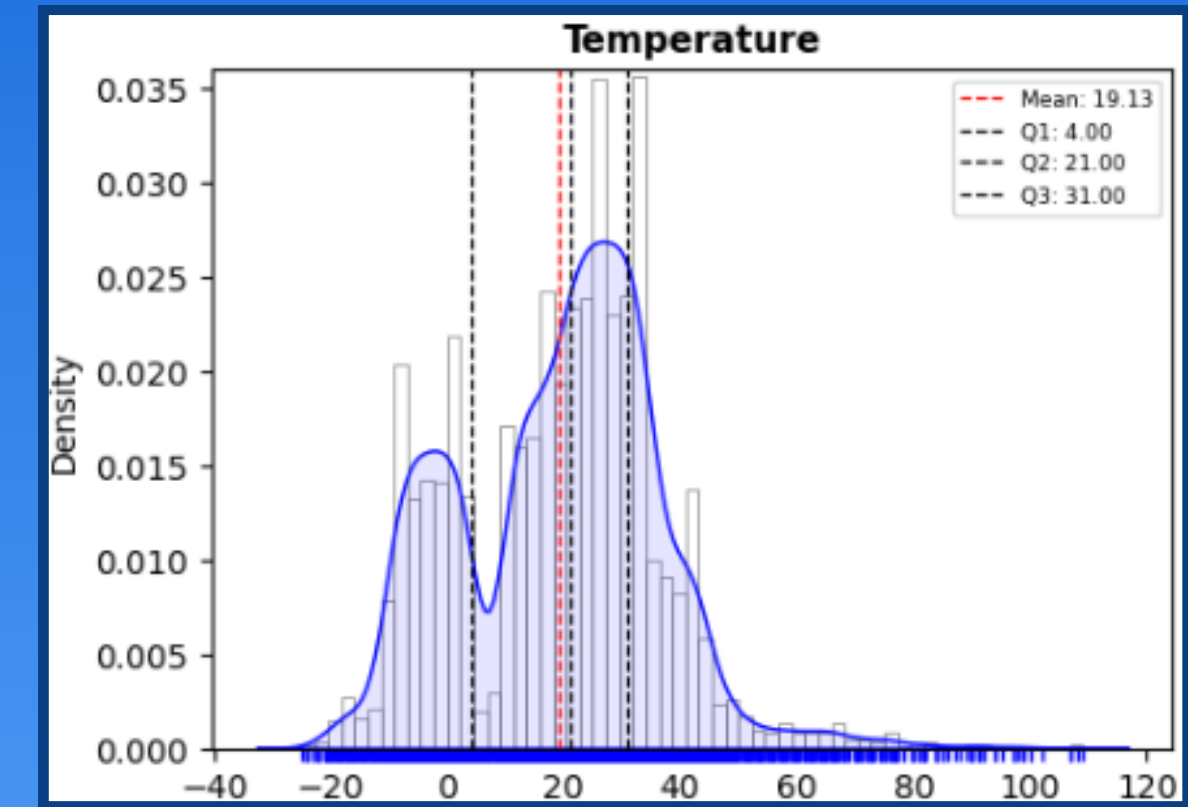


- Cloud Cover
- Season
- Location
- Weather Type (Target)

📍 Outlier Detection

Outliers Detected (IQR Method):

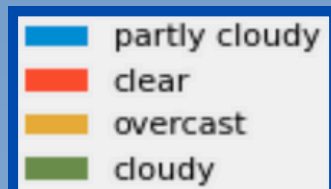
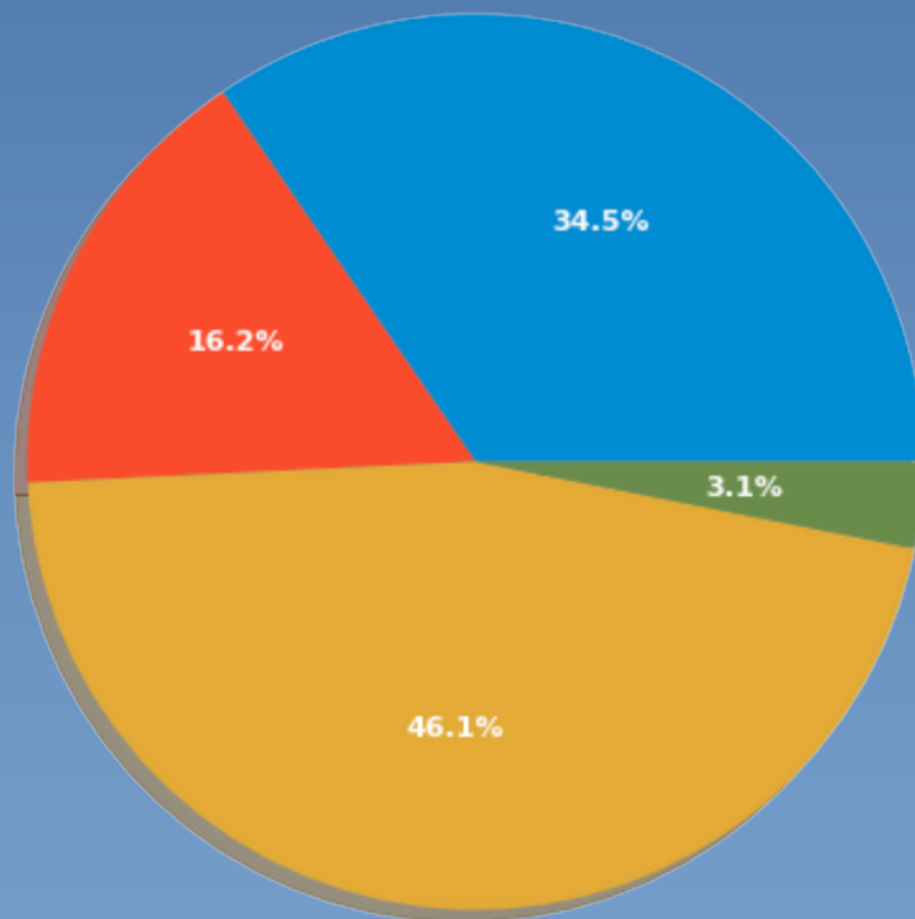
- Temperature: 92
- Wind Speed: 404
- Atmospheric Pressure: 927
- Visibility (km): 383
- Humidity, Precipitation, UV Index: No significant outliers detected.



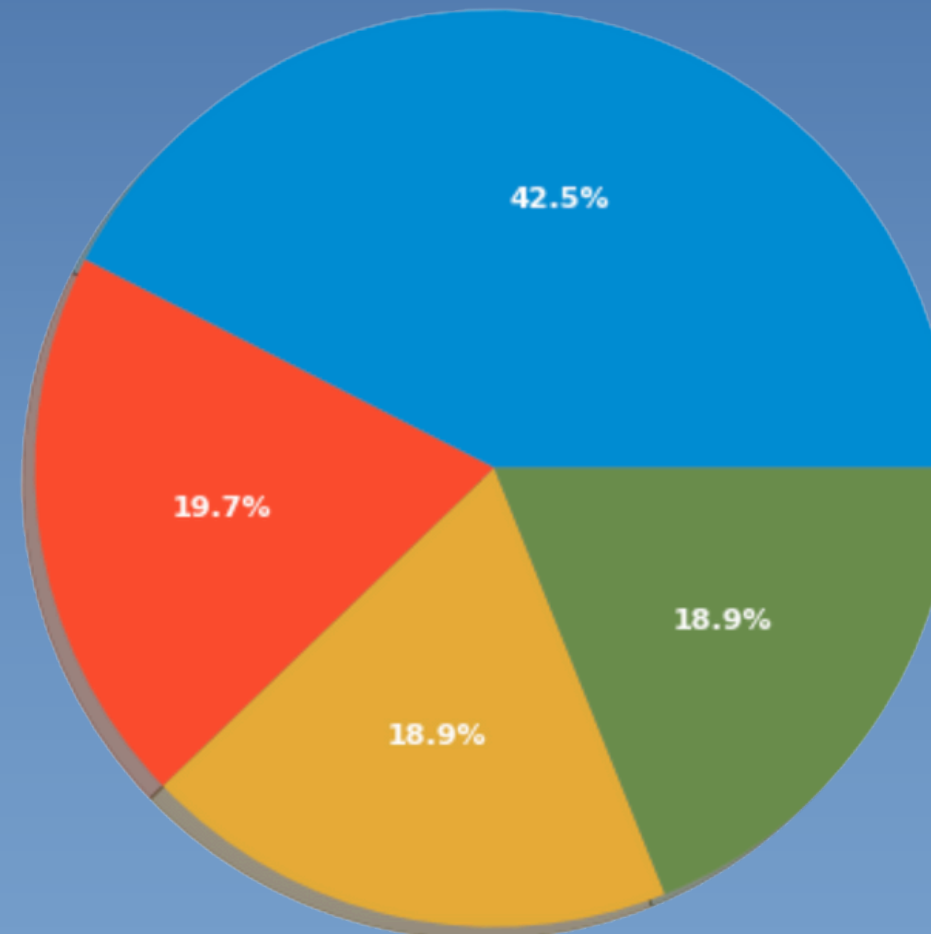
📍 Categorical Feature Distributions



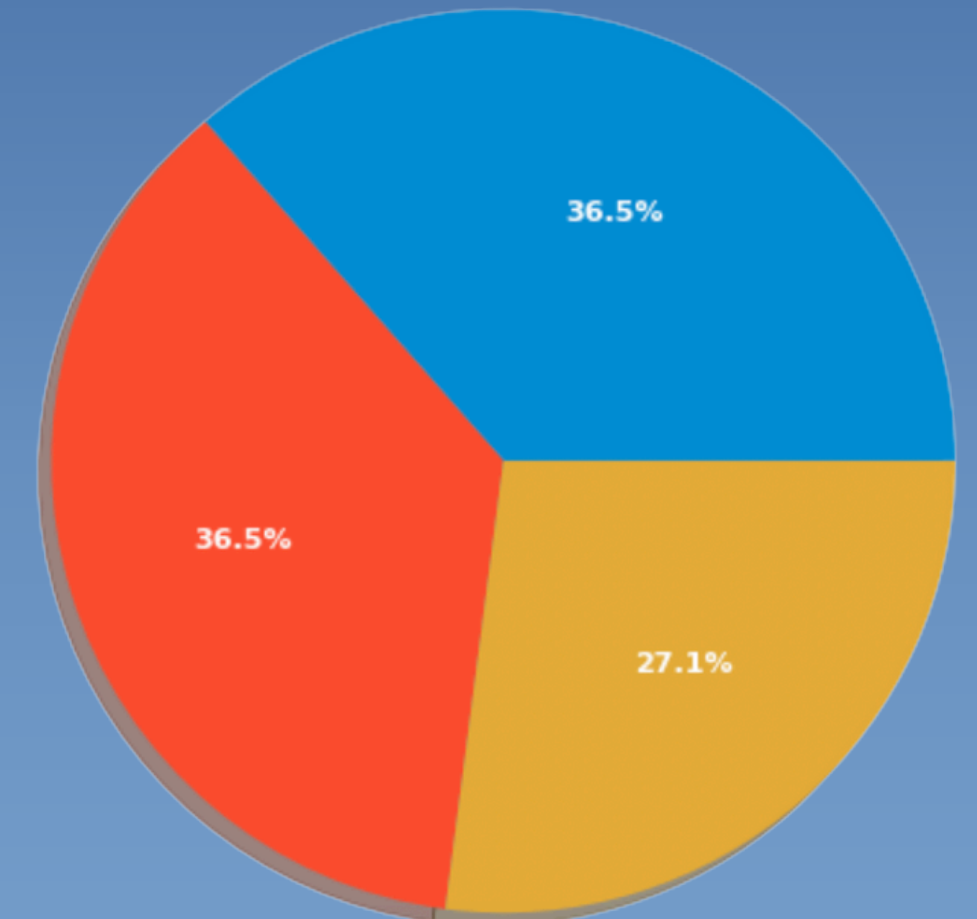
Cloud Cover



Season



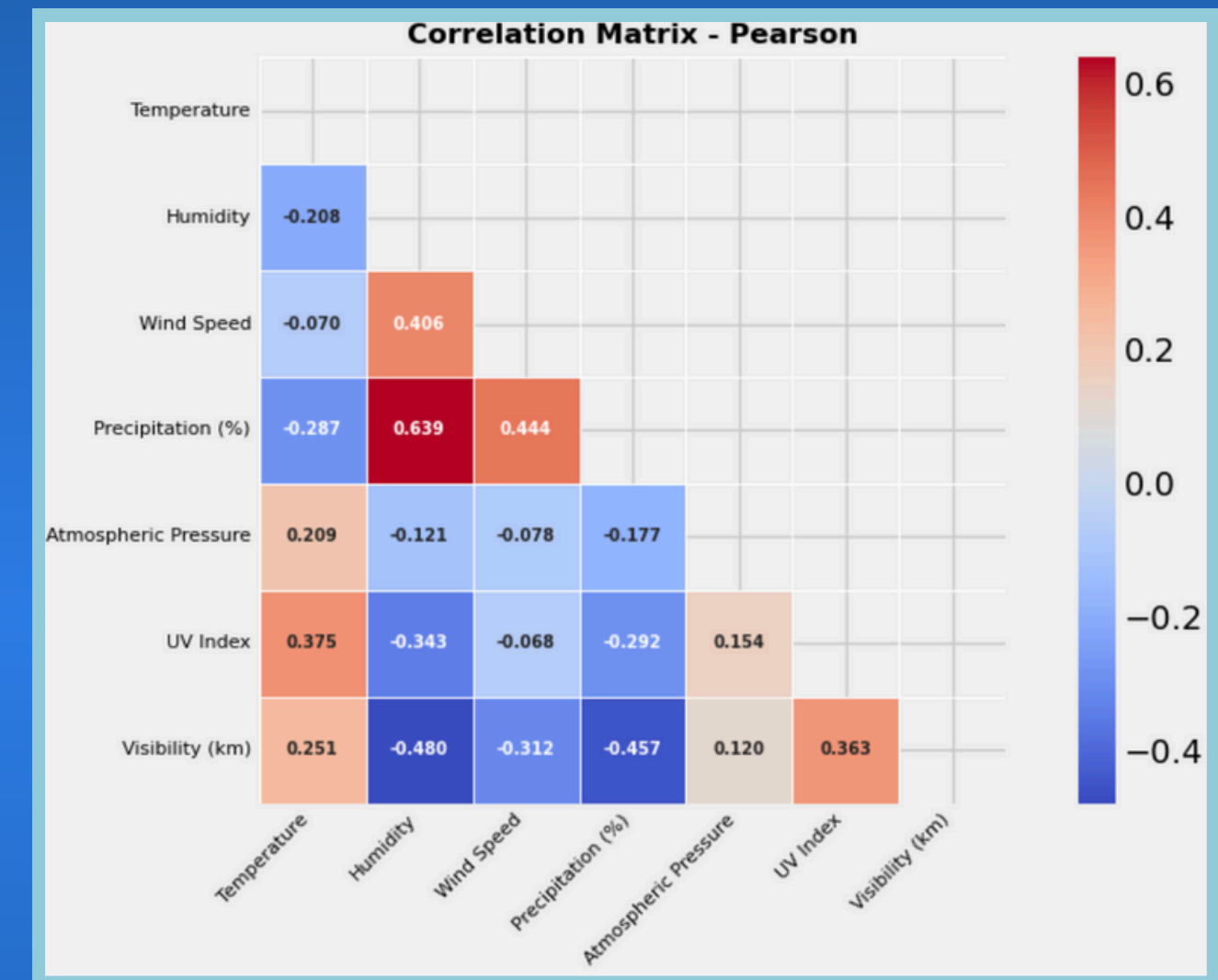
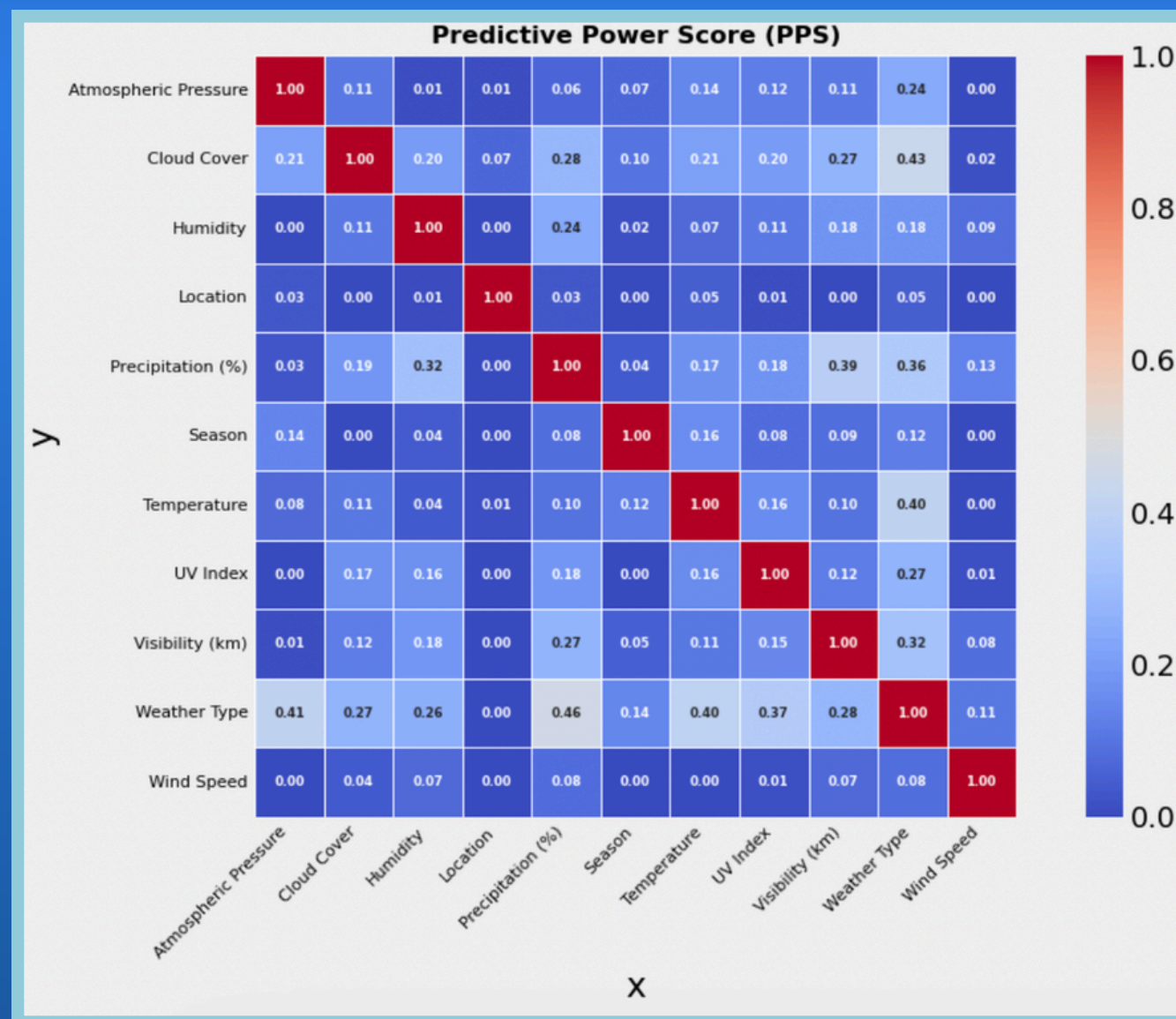
Location



📍 Feature Relationship Matrices

Correlation Analysis:

- Strong correlation: Humidity & Precipitation (0.639)
- Moderate correlation: Wind Speed & Precipitation (0.444)



Predictive Power Score (PPS) Analysis:

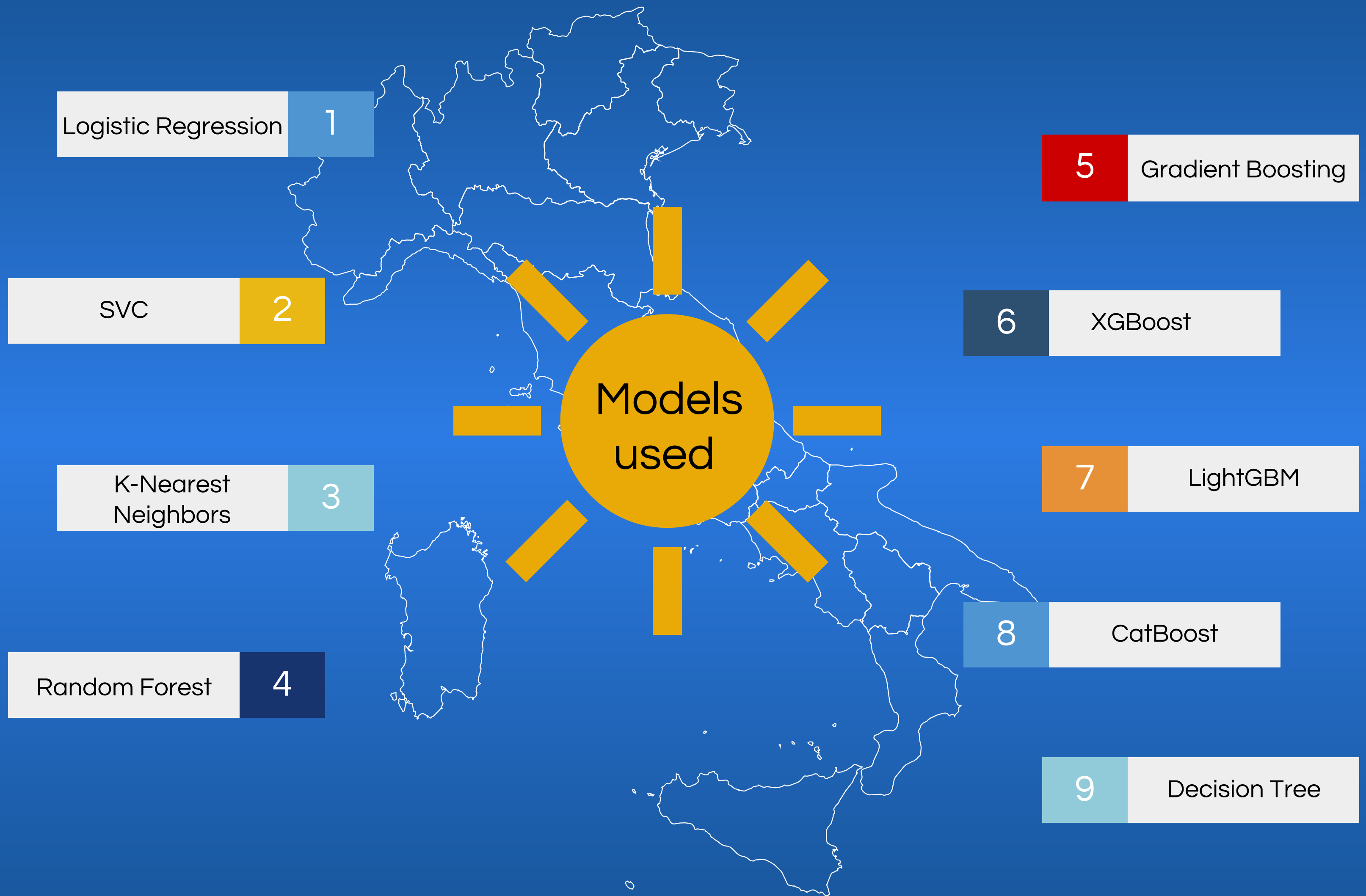
Top predictors of Weather Type:

- Precipitation (0.46), Atmospheric Pressure (0.41), Temperature (0.40)

📍 Data Preparation

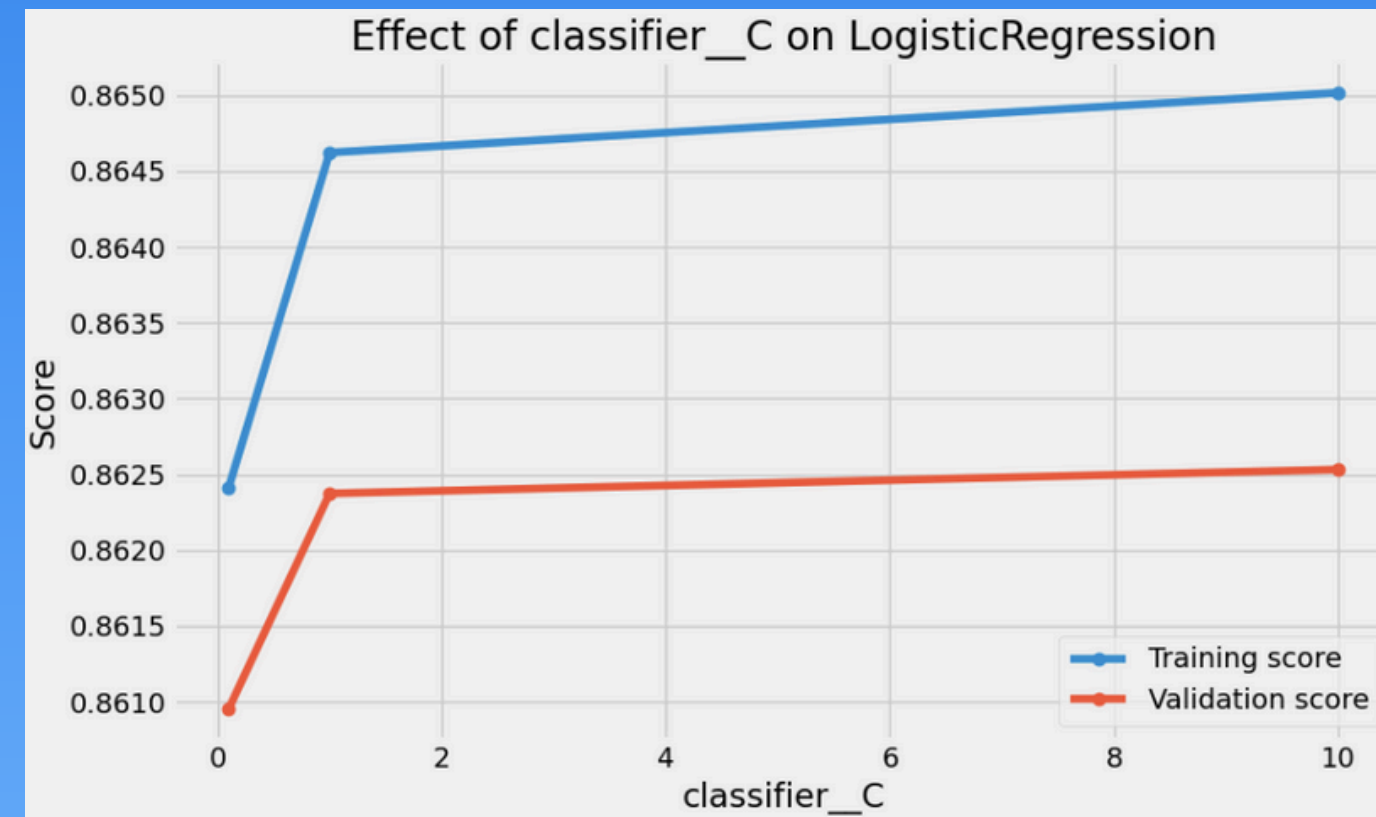
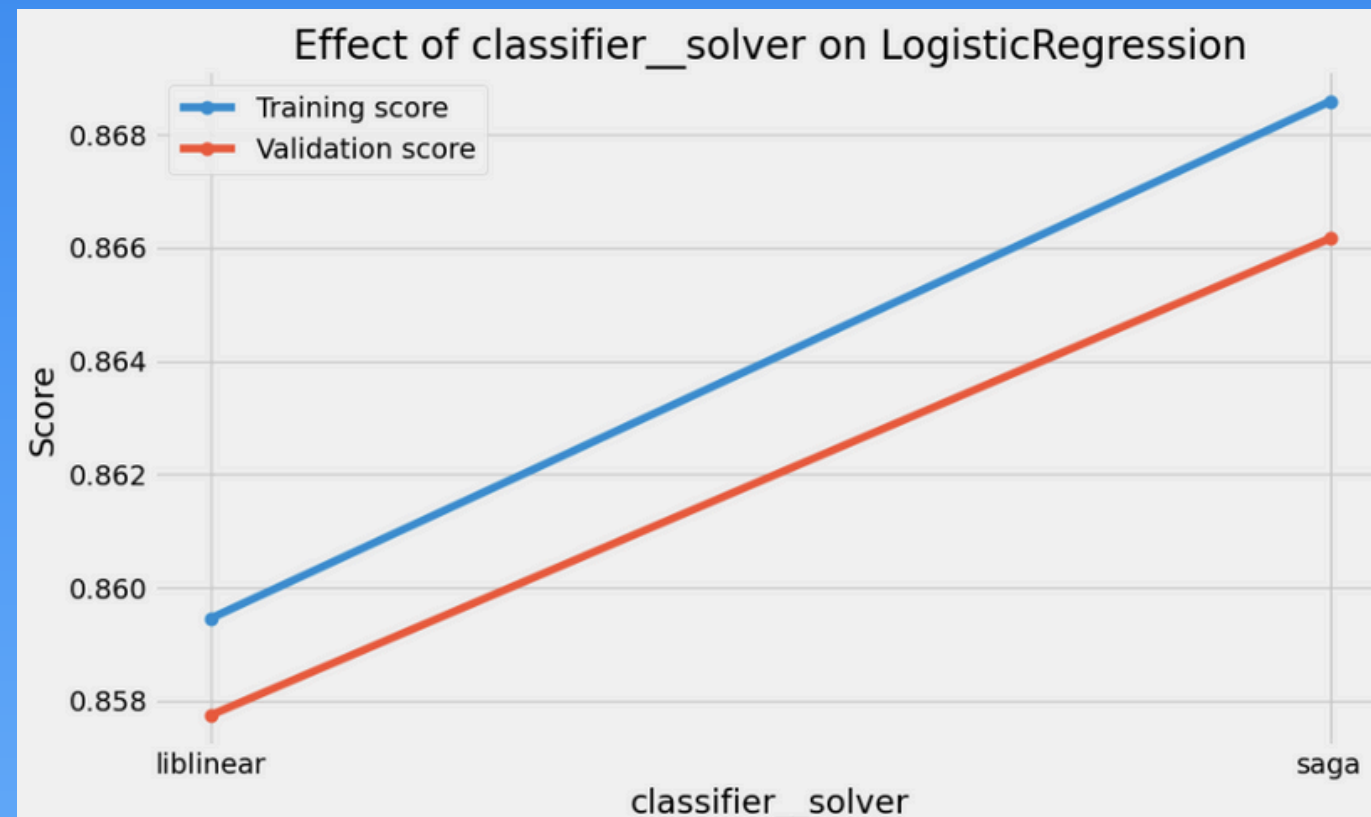
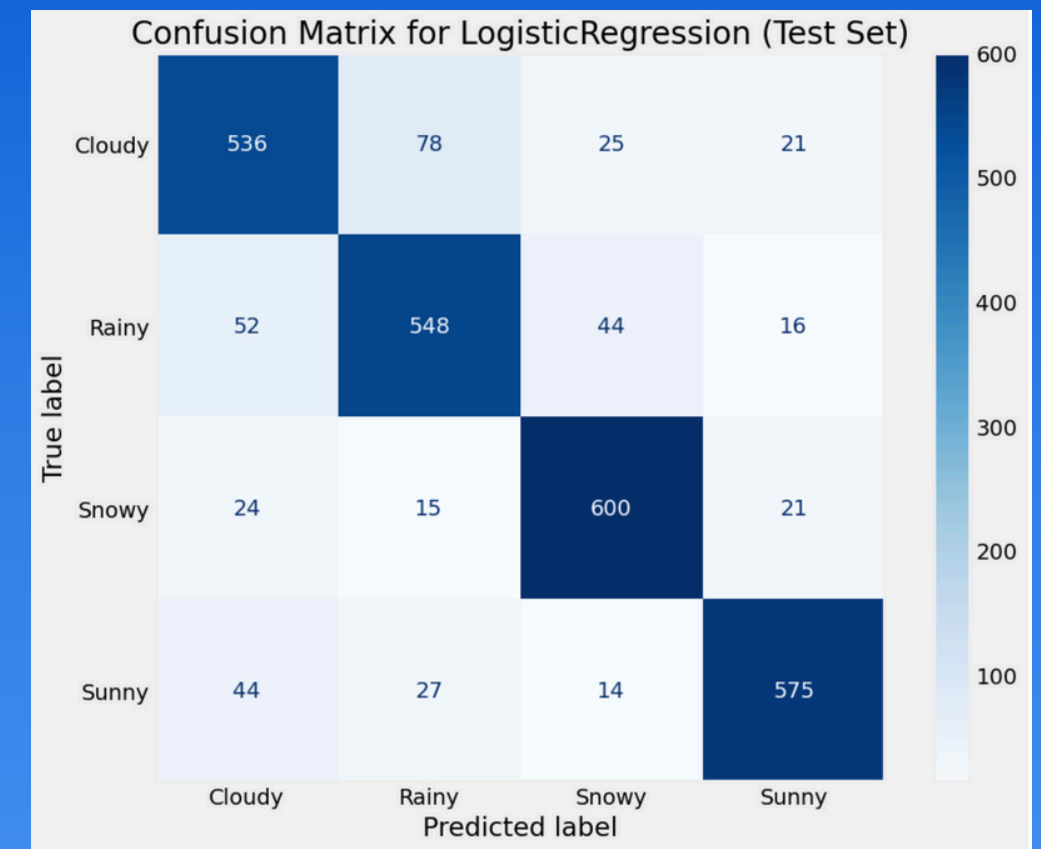


- **Data Splitting:** The data is split into training (72%), validation (8%), and testing (20%) sets.
- **Feature Preprocessing:** Numerical features are scaled using `StandardScaler`. Categorical features are one-hot encoded using `OneHotEncoder`.
- **Class Balancing:** SMOTE (Synthetic Minority Oversampling Technique) is used to oversample the minority classes in the training set.
- **Label Encoding:** `LabelEncoder` is used to convert the string labels of the "Weather Type" into numerical representations.

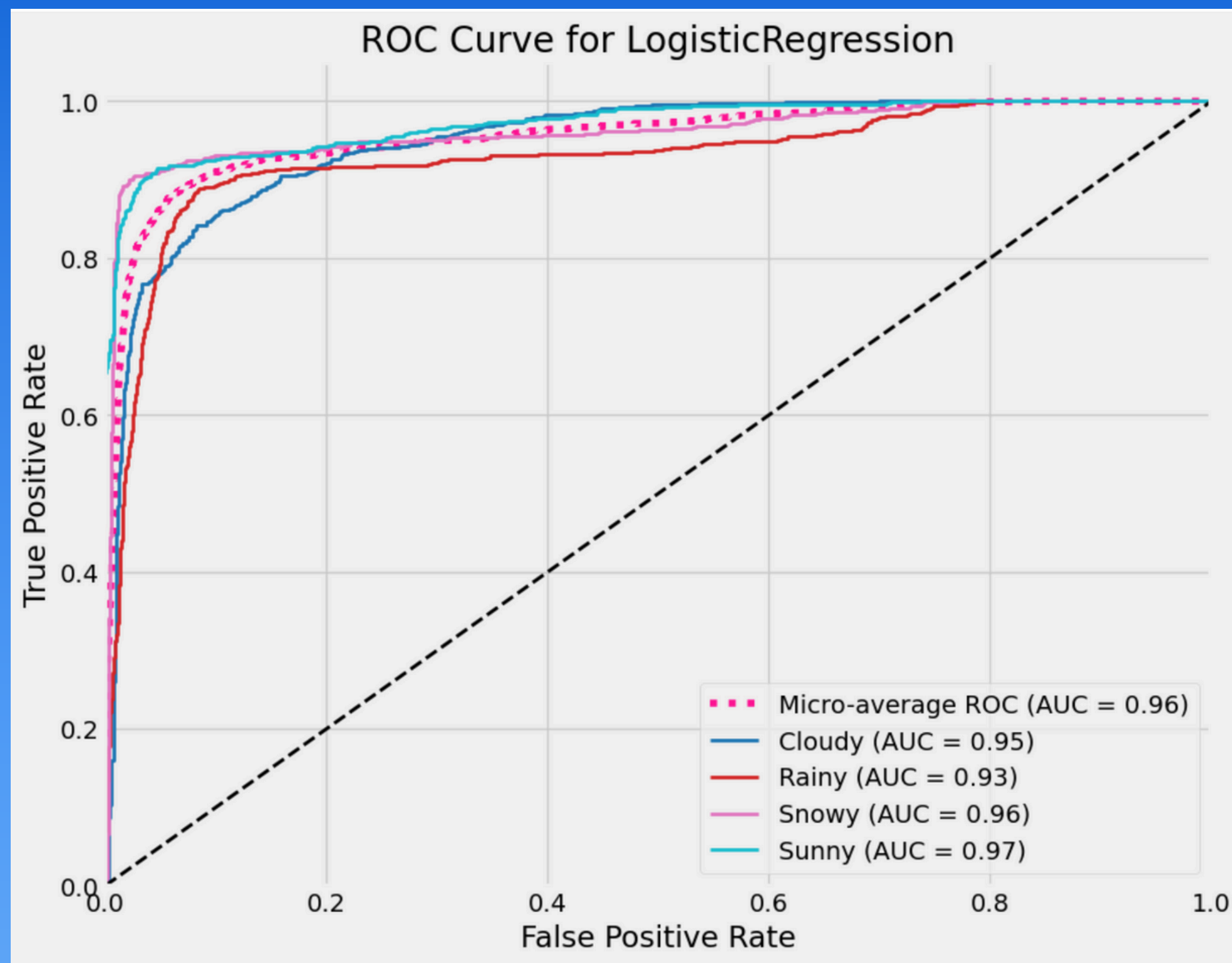


Hyperparameters Used:

- Regularization Parameter (C): 10 (higher C improved performance).
- Solver: 'saga' (worked better than 'liblinear' for optimization).



Logistic Regression



Classification Report for LogisticRegression (Test Set):				
	precision	recall	f1-score	support
Cloudy	0.82	0.81	0.81	660
Rainy	0.82	0.83	0.83	660
Snowy	0.88	0.91	0.89	660
Sunny	0.91	0.87	0.89	660
accuracy			0.86	2640
macro avg	0.86	0.86	0.86	2640
weighted avg	0.86	0.86	0.86	2640

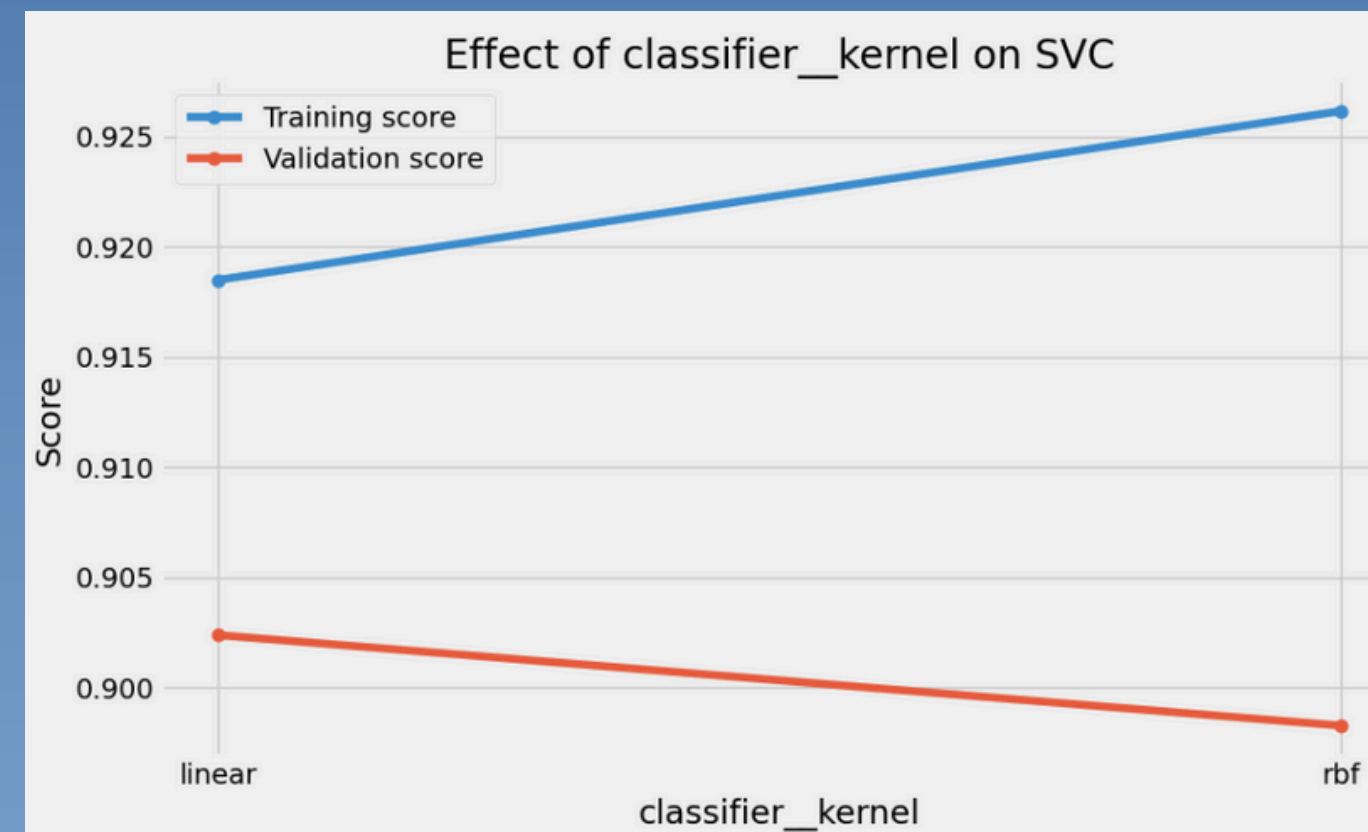
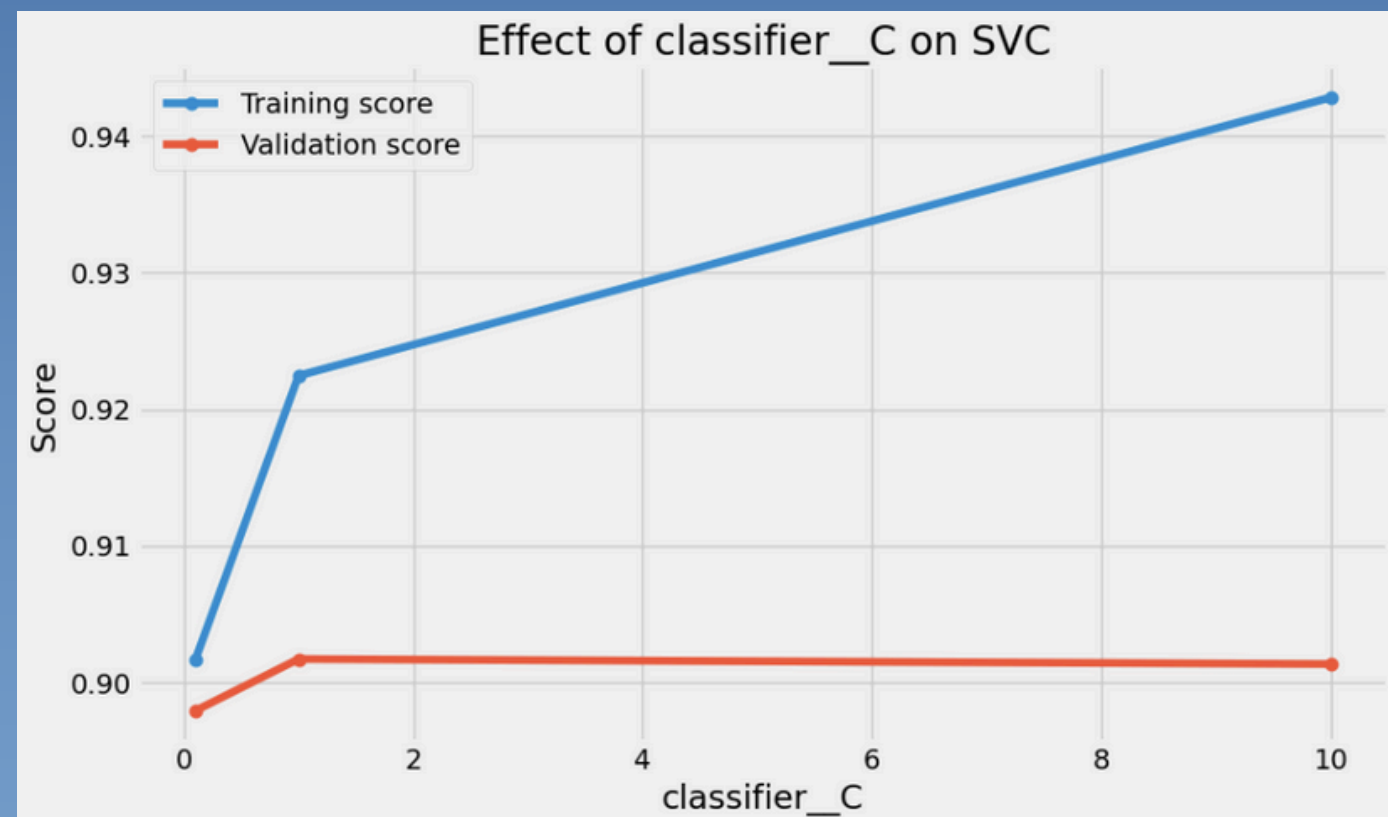
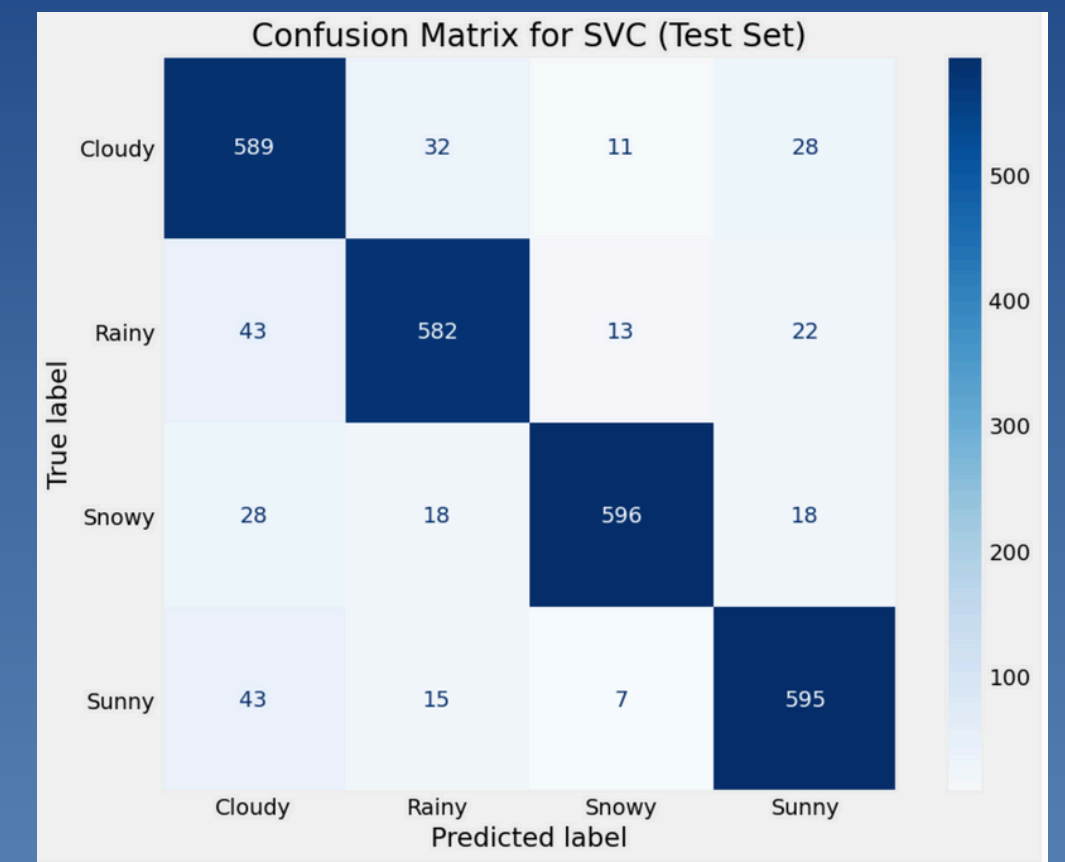
Performance Metrics:

- Accuracy: 86% (good overall performance).
- Precision: 0.82 - 0.91 (model correctly identifies positive cases).
- Recall: 0.81 - 0.91 (ability to capture all positive cases).
- F1-score: 0.83 - 0.89 (balance between precision and recall).

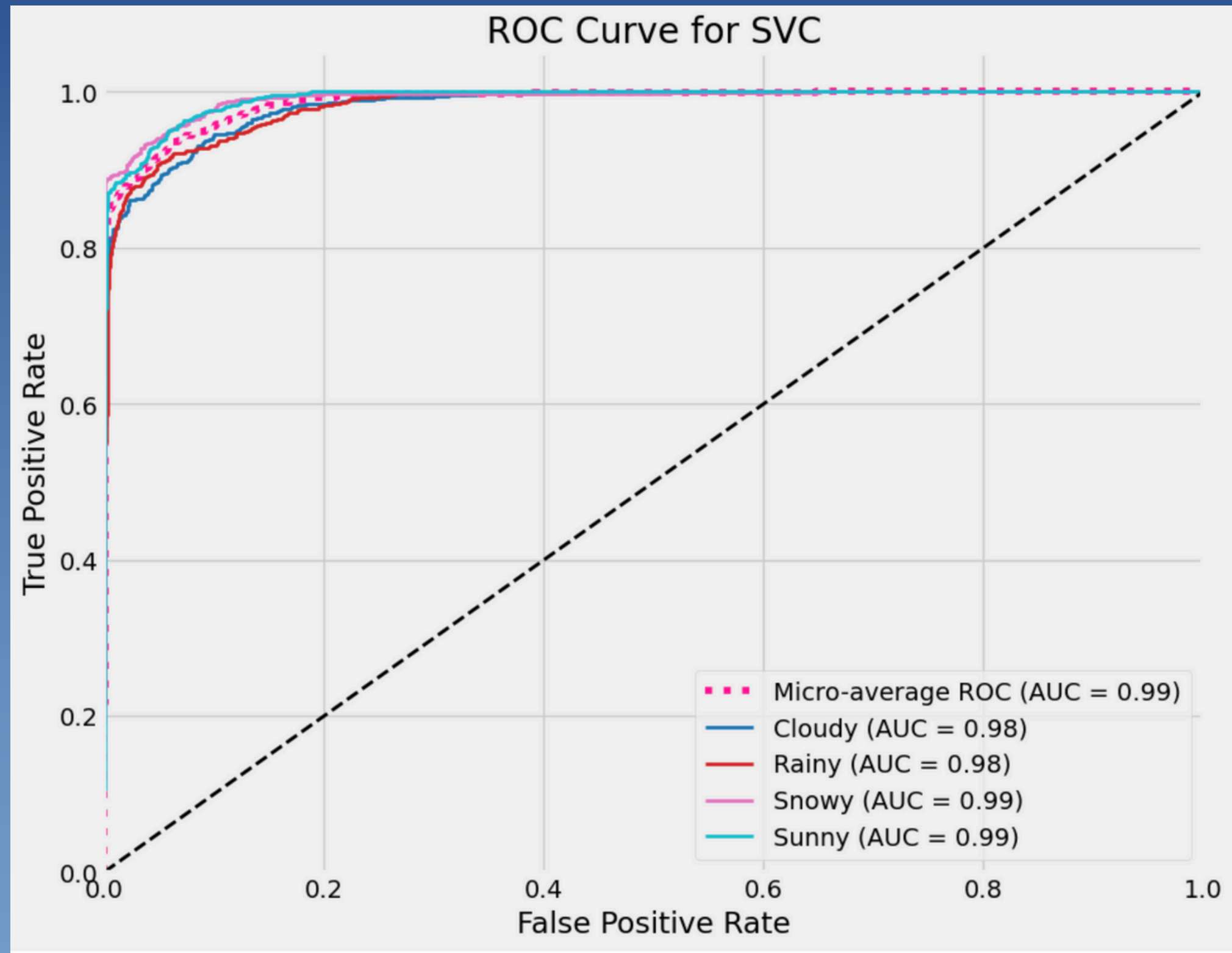
Logistic Regression

Hyperparameters Used:

- Regularization Parameter (C): 1 (controls the margin size and complexity).
- Kernel: RBF (captures non-linear relationships in data).



SVC



Classification Report for SVC (Test Set):				
	precision	recall	f1-score	support
Cloudy	0.84	0.89	0.86	660
Rainy	0.90	0.88	0.89	660
Snowy	0.95	0.90	0.93	660
Sunny	0.90	0.90	0.90	660
accuracy			0.89	2640
macro avg	0.90	0.89	0.90	2640
weighted avg	0.90	0.89	0.90	2640

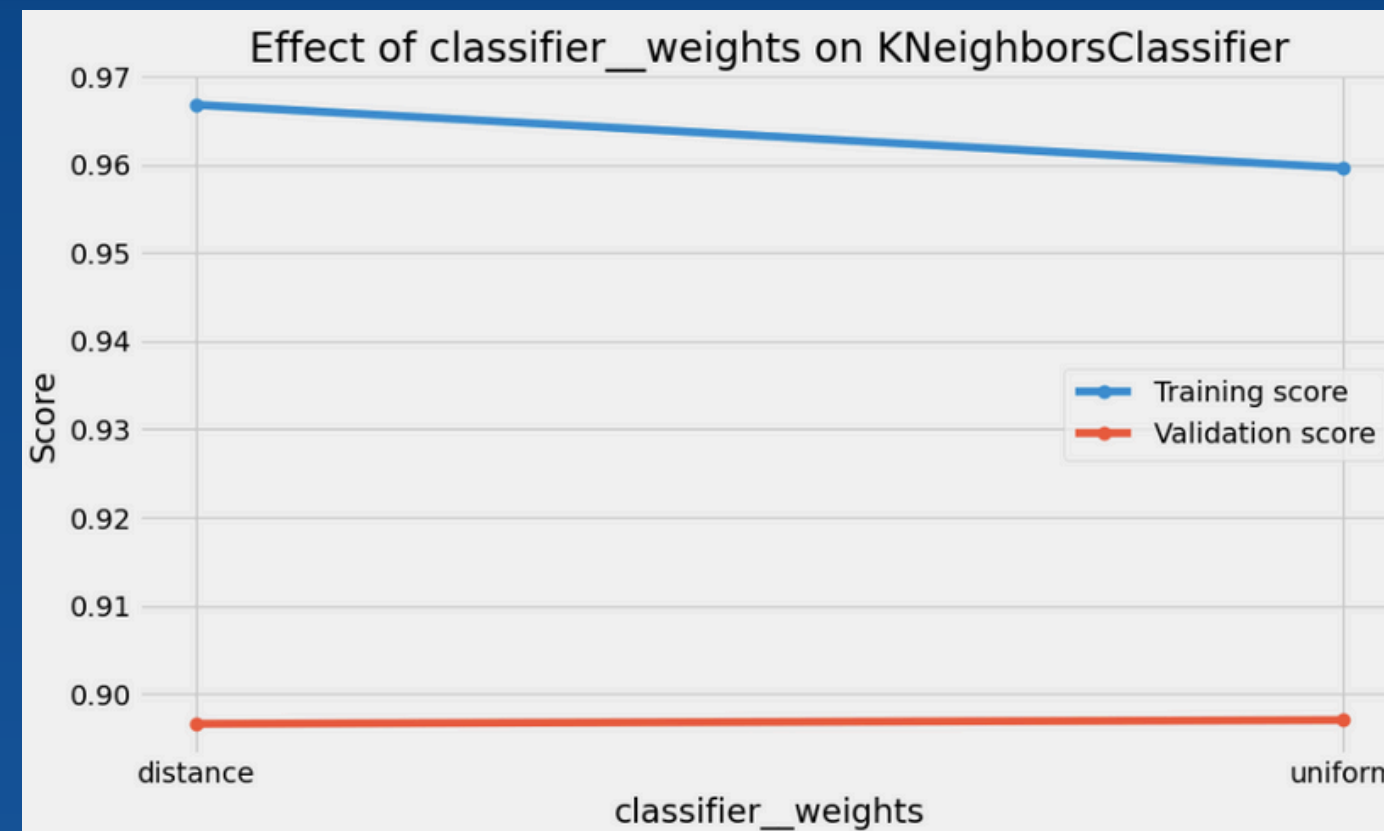
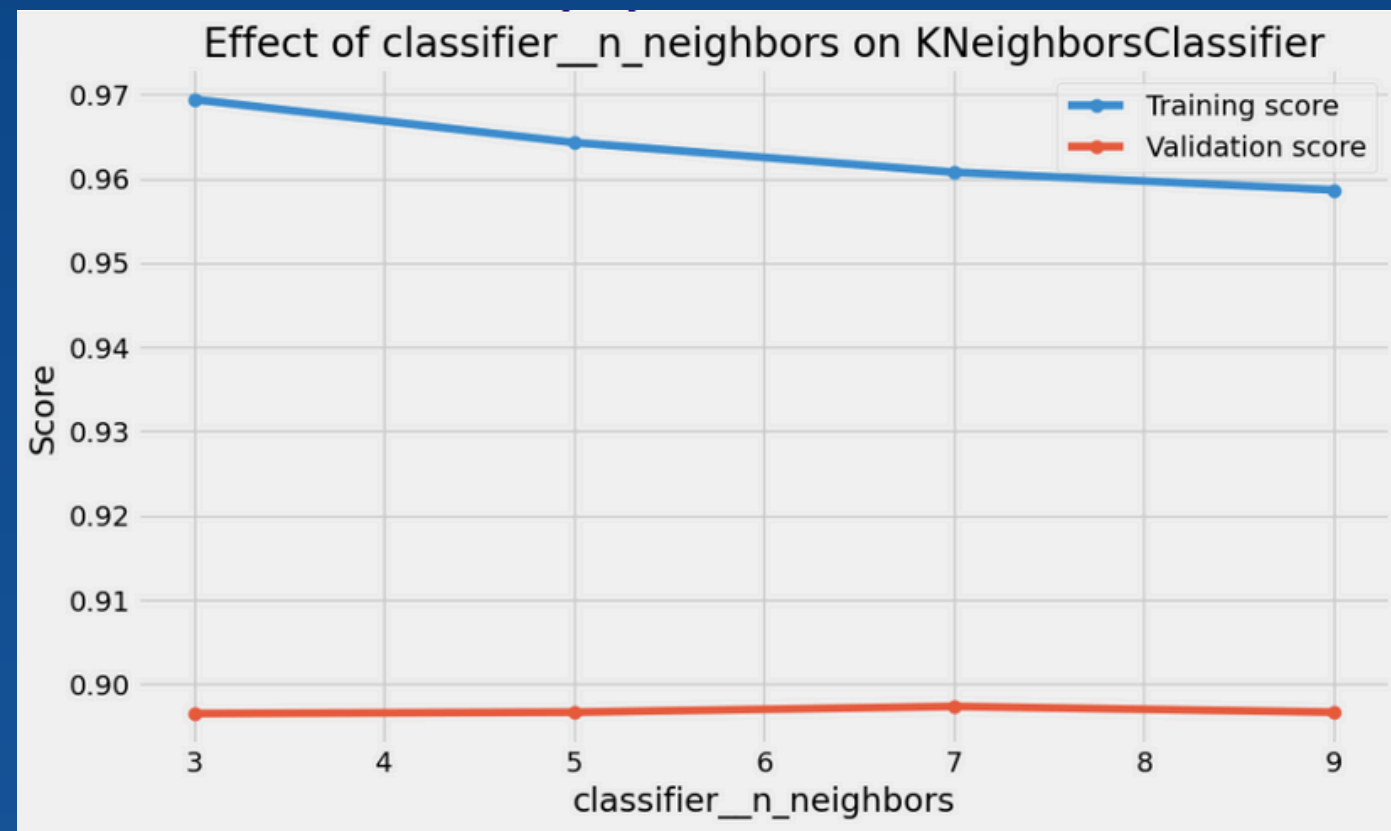
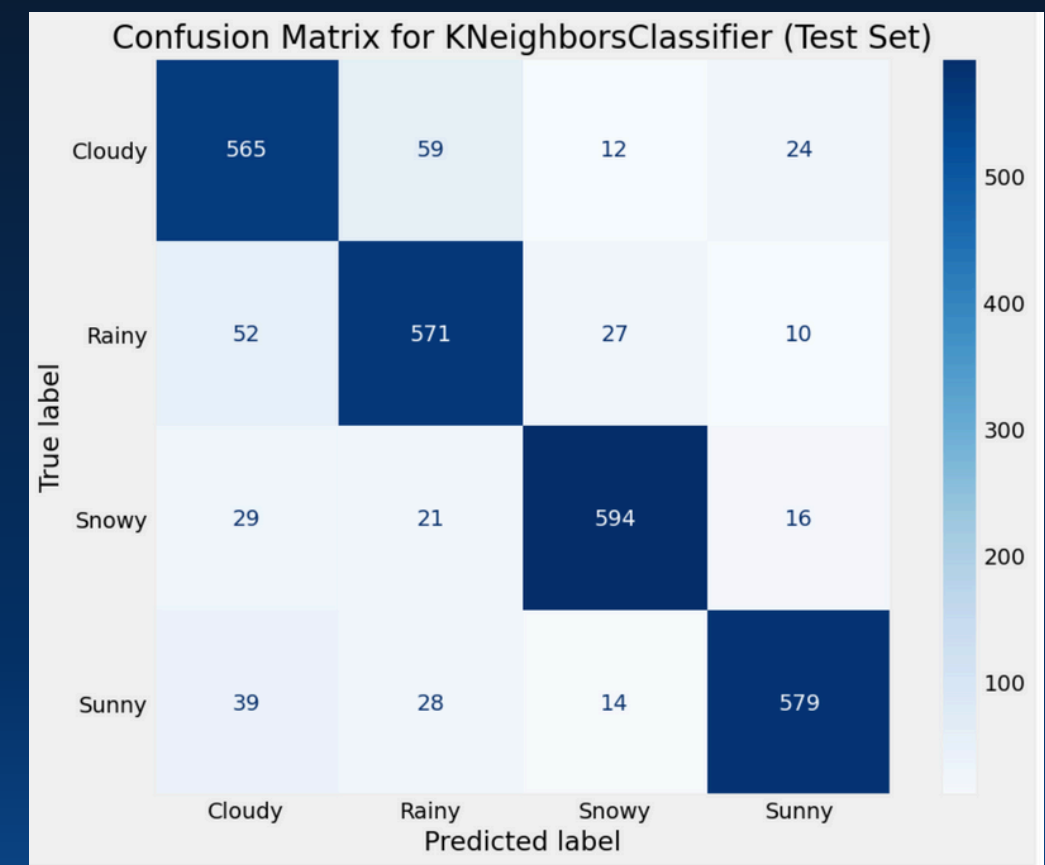
Performance Metrics:

- Accuracy: 89% (better overall performance than Logistic Regression).
- Precision: 0.84 - 0.95 (model correctly identifies positive cases).
- Recall: 0.88 - 0.90 (ability to capture all positive cases).
- F1-score: 0.86 - 0.93 (balance between precision and recall)

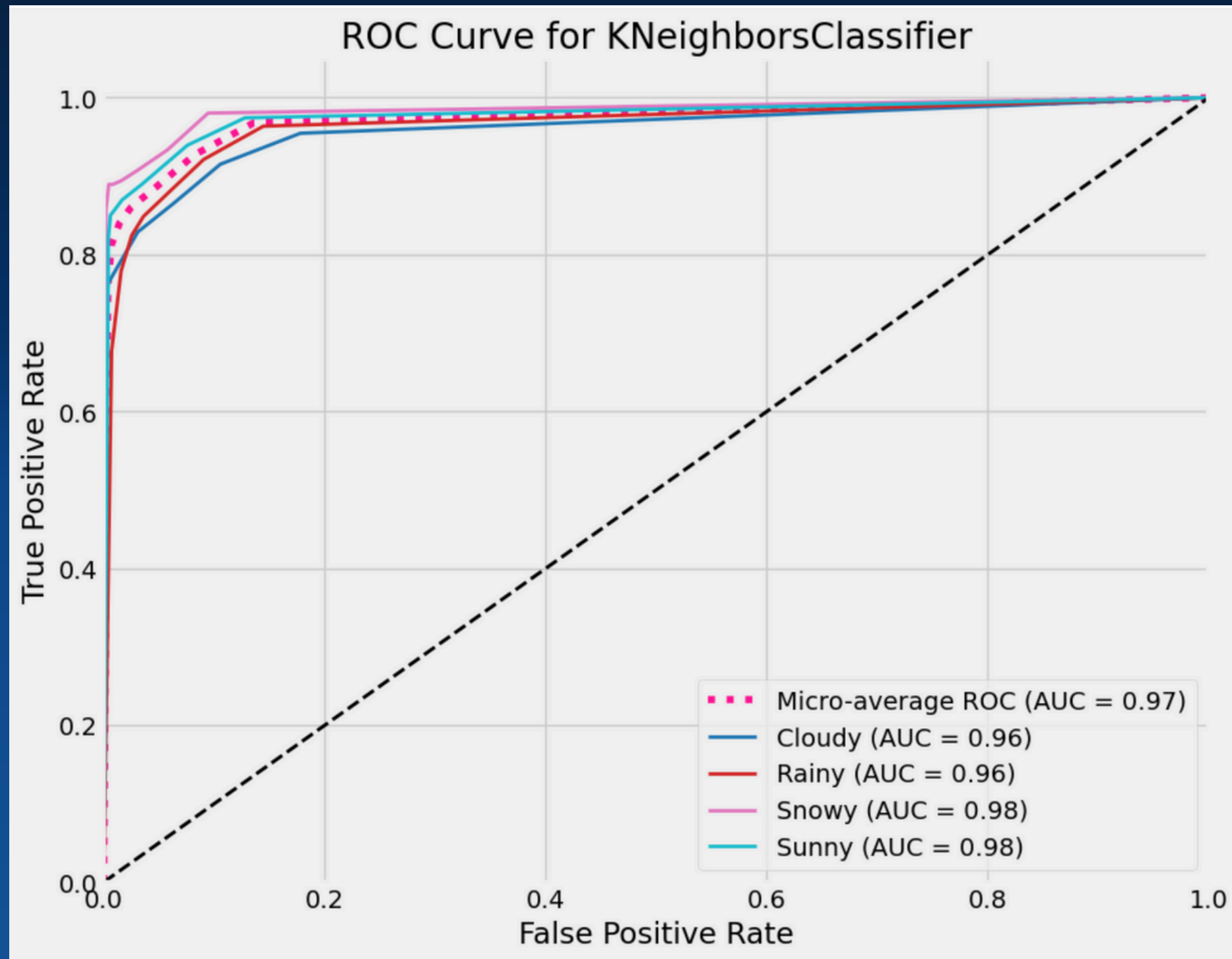
SVC

Hyperparameters Used:

- Number of Neighbors (k): 7 (determines how many nearest points influence classification).
- Weights: 'uniform' (all neighbors contribute equally to the classification).



KNN Classifier



Classification Report for KNeighborsClassifier (Test Set):				
	precision	recall	f1-score	support
Cloudy	0.82	0.86	0.84	660
Rainy	0.84	0.87	0.85	660
Snowy	0.92	0.90	0.91	660
Sunny	0.92	0.88	0.90	660
accuracy			0.87	2640
macro avg	0.88	0.87	0.88	2640
weighted avg	0.88	0.87	0.88	2640

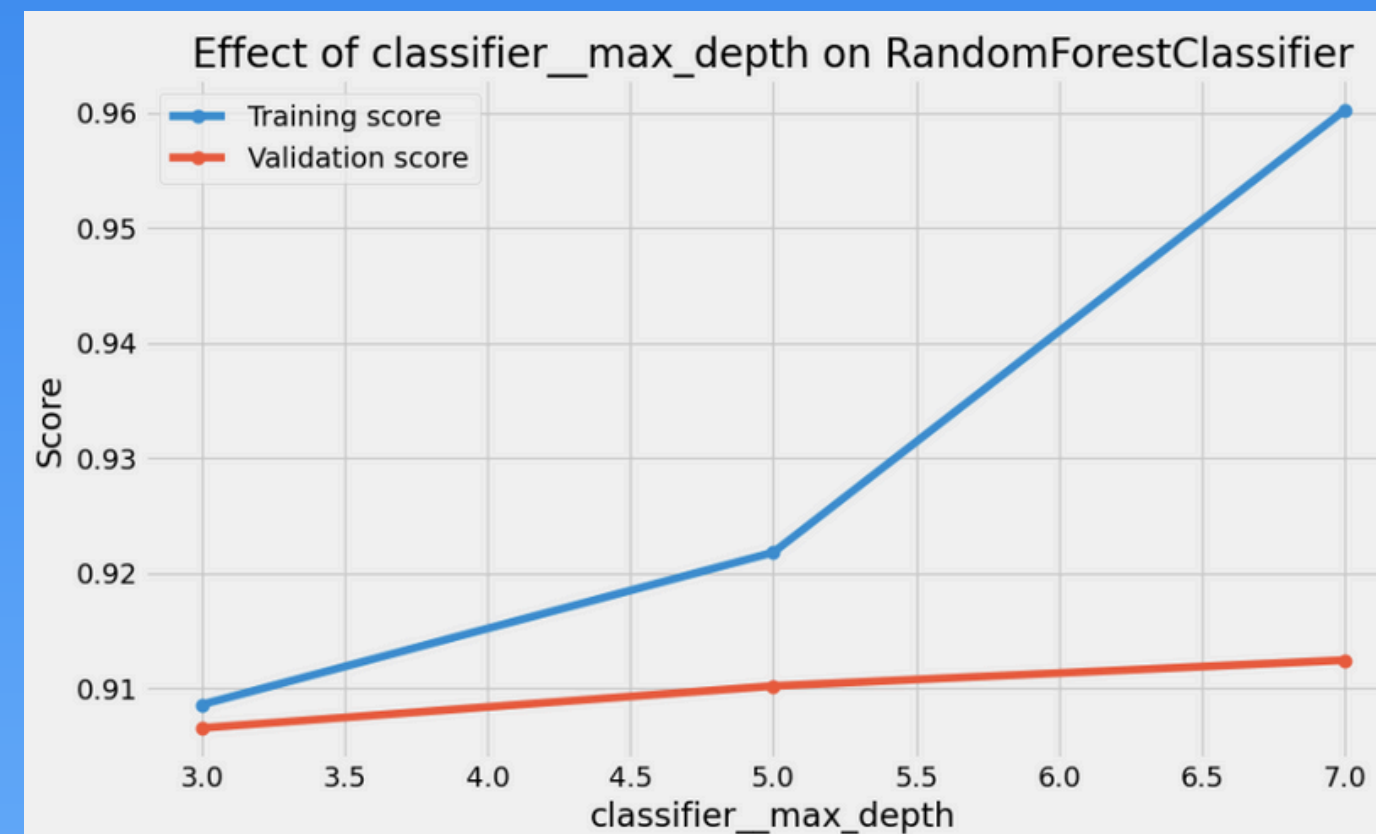
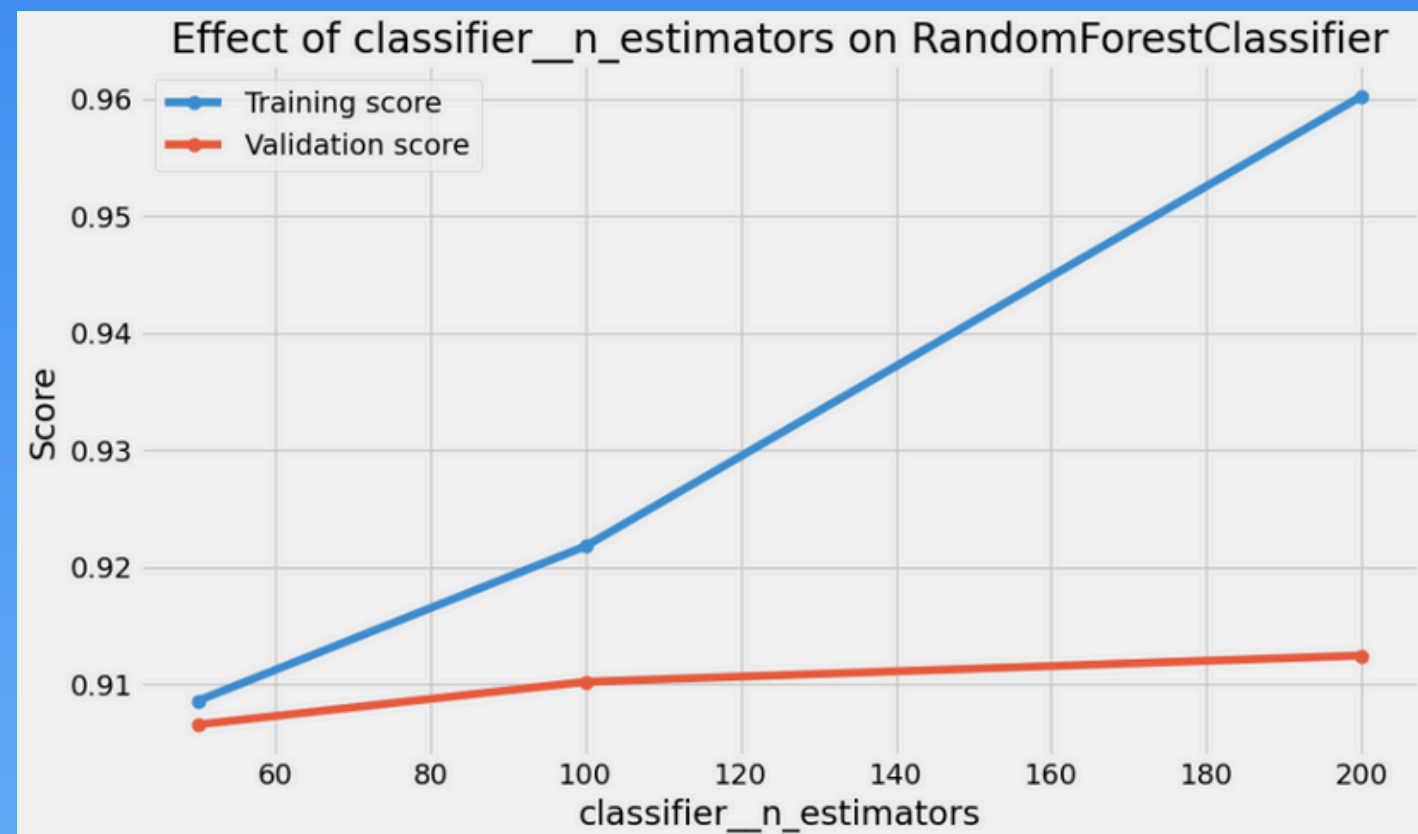
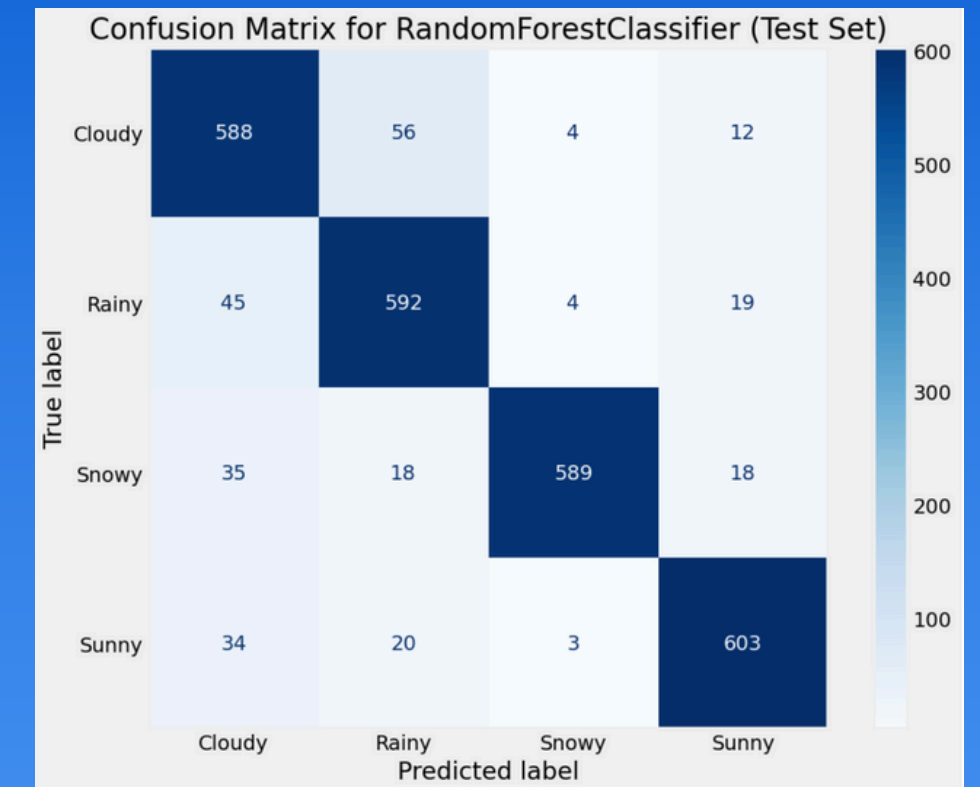
Performance Metrics:

- Accuracy: 87% (slightly lower than tree-based models).
- Precision: 0.82 - 0.92 (model correctly identifies positive cases).
- Recall: 0.86 - 90 (ability to capture all positive cases).
- F1-score: 0.84 - 0.91 (balance between precision and recall).

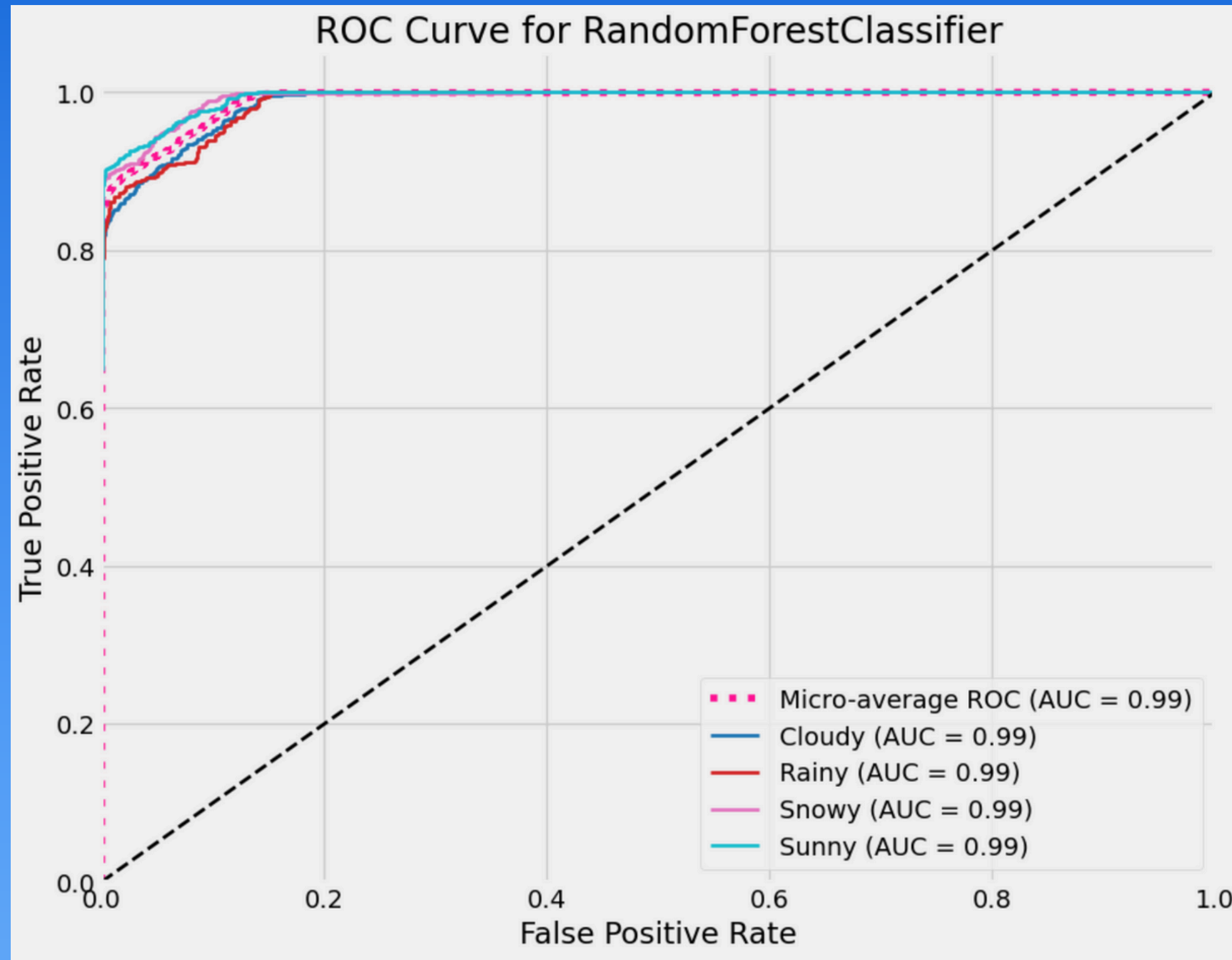
KNN Classifier

Hyperparameters Used:

- Number of Estimators (Trees): 100 (higher trees improve stability).
- Maximum Depth: 7 (limits tree depth to prevent overfitting).



Random Forest Classifier



Classification Report for RandomForestClassifier (Test Set):				
	precision	recall	f1-score	support
Cloudy	0.84	0.89	0.86	660
Rainy	0.86	0.90	0.88	660
Snowy	0.98	0.89	0.93	660
Sunny	0.92	0.91	0.92	660
accuracy			0.90	2640
macro avg	0.90	0.90	0.90	2640
weighted avg	0.90	0.90	0.90	2640

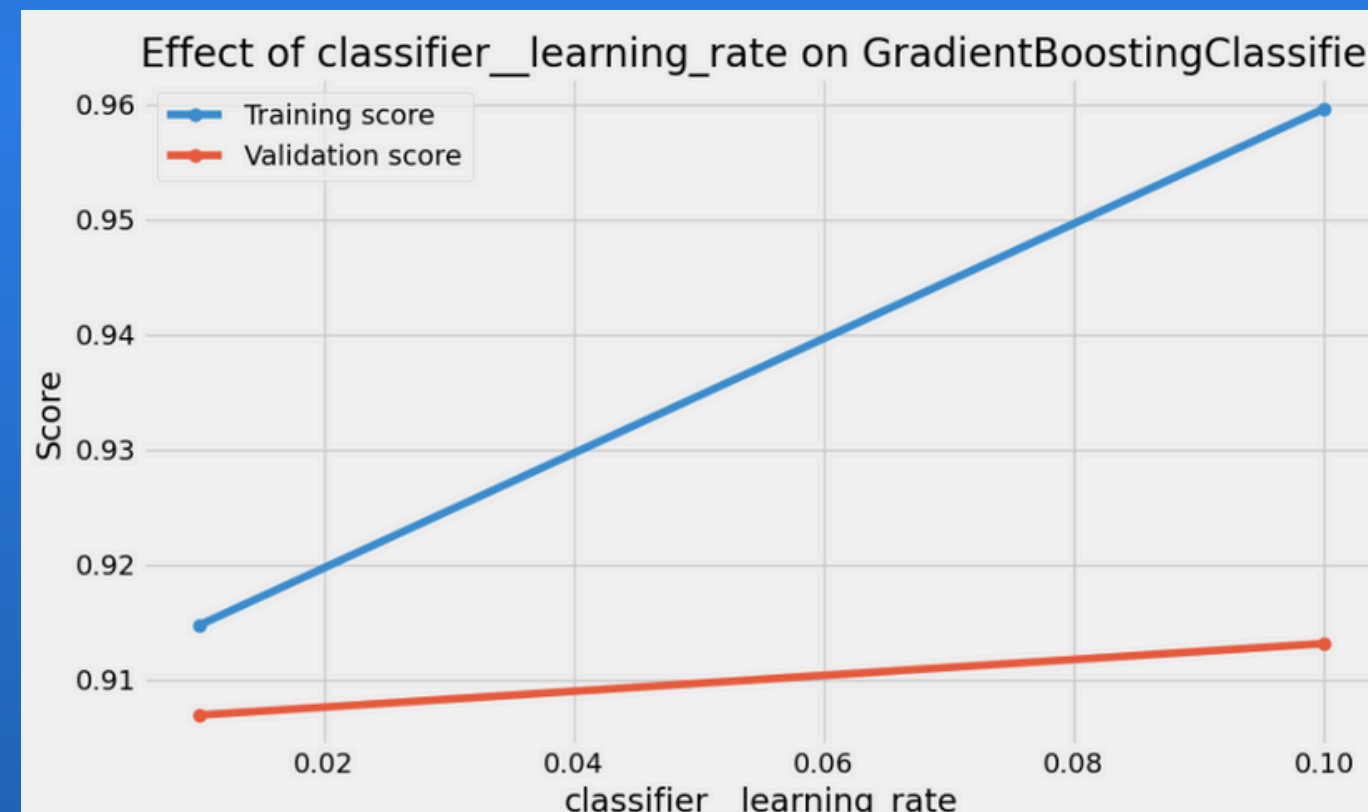
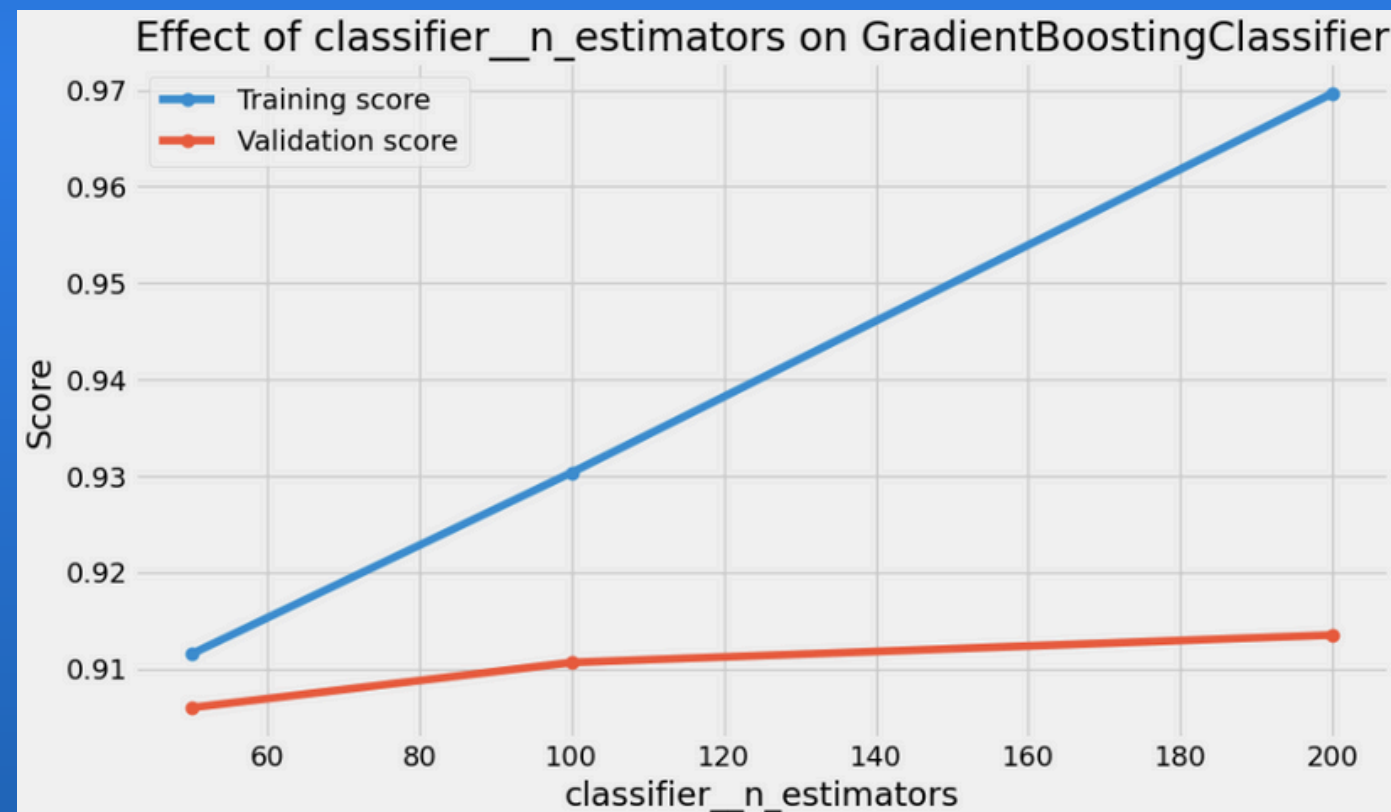
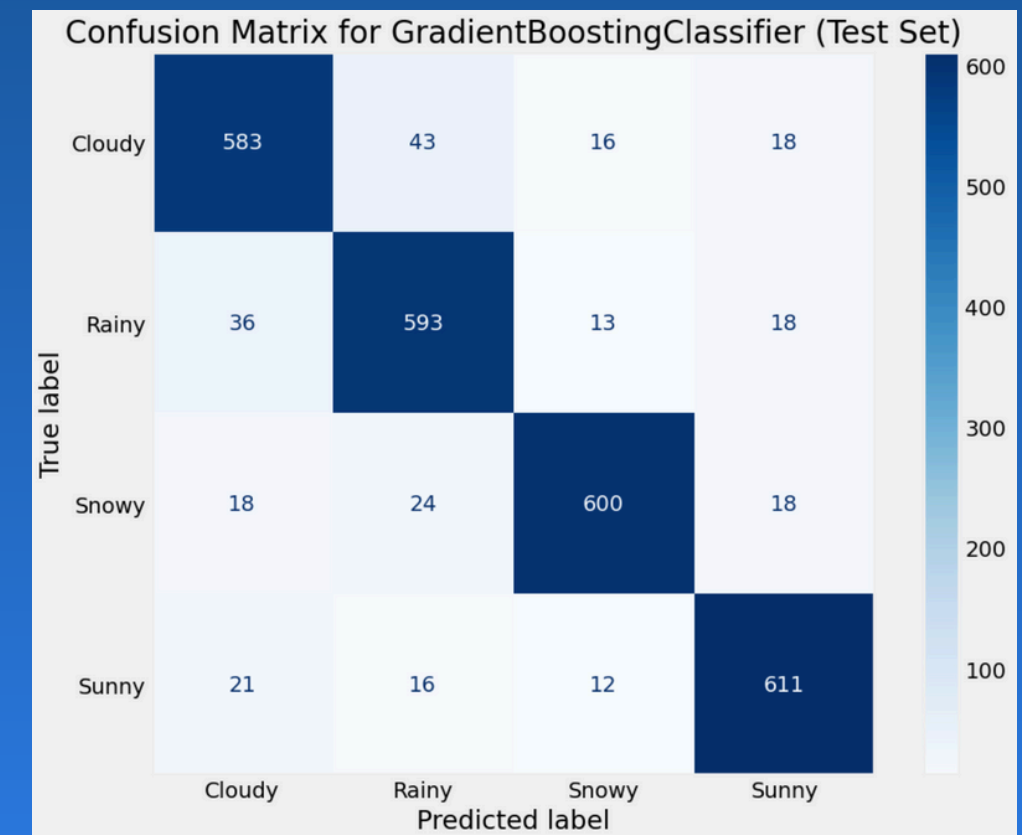
Performance Metrics:

- Accuracy: 90% (best overall performance so far).
- Precision: 0.84 - 0.98 (model correctly identifies positive cases).
- Recall: 0.89 - 0.91 (ability to capture all positive cases).
- F1-score: 0.86 - 0.93 (balance between precision and recall).

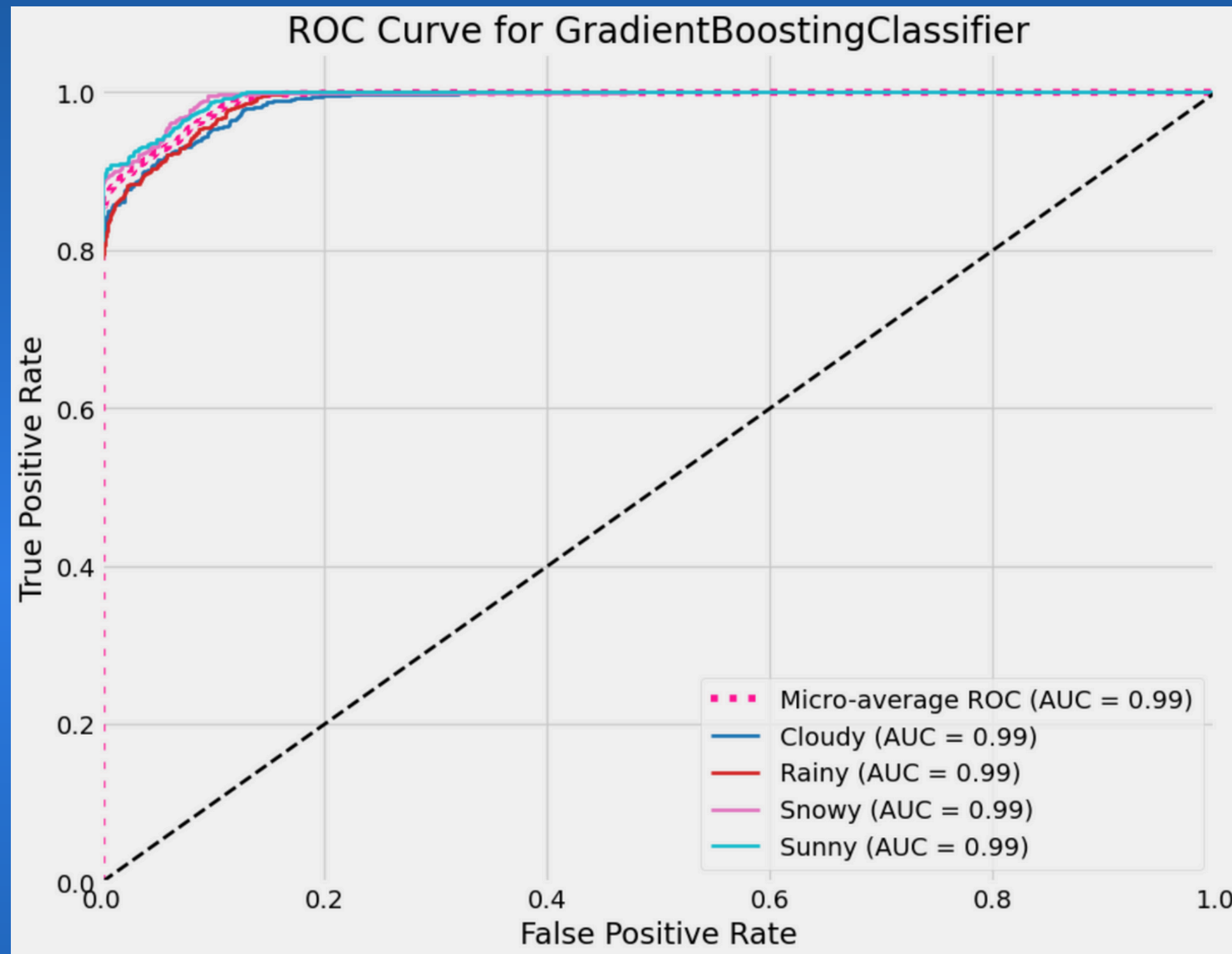
Random Forest Classifier

Hyperparameters Used:

- Learning Rate: 0.1 (controls step size during optimization).
- Number of Estimators: 100 (defines the number of boosting stages).



Gradient Boosting Classifier



Classification Report for GradientBoostingClassifier (Test Set):				
	precision	recall	f1-score	support
Cloudy	0.89	0.88	0.88	660
Rainy	0.88	0.90	0.89	660
Snowy	0.94	0.91	0.92	660
Sunny	0.92	0.93	0.92	660
accuracy			0.90	2640
macro avg	0.90	0.90	0.90	2640
weighted avg	0.90	0.90	0.90	2640

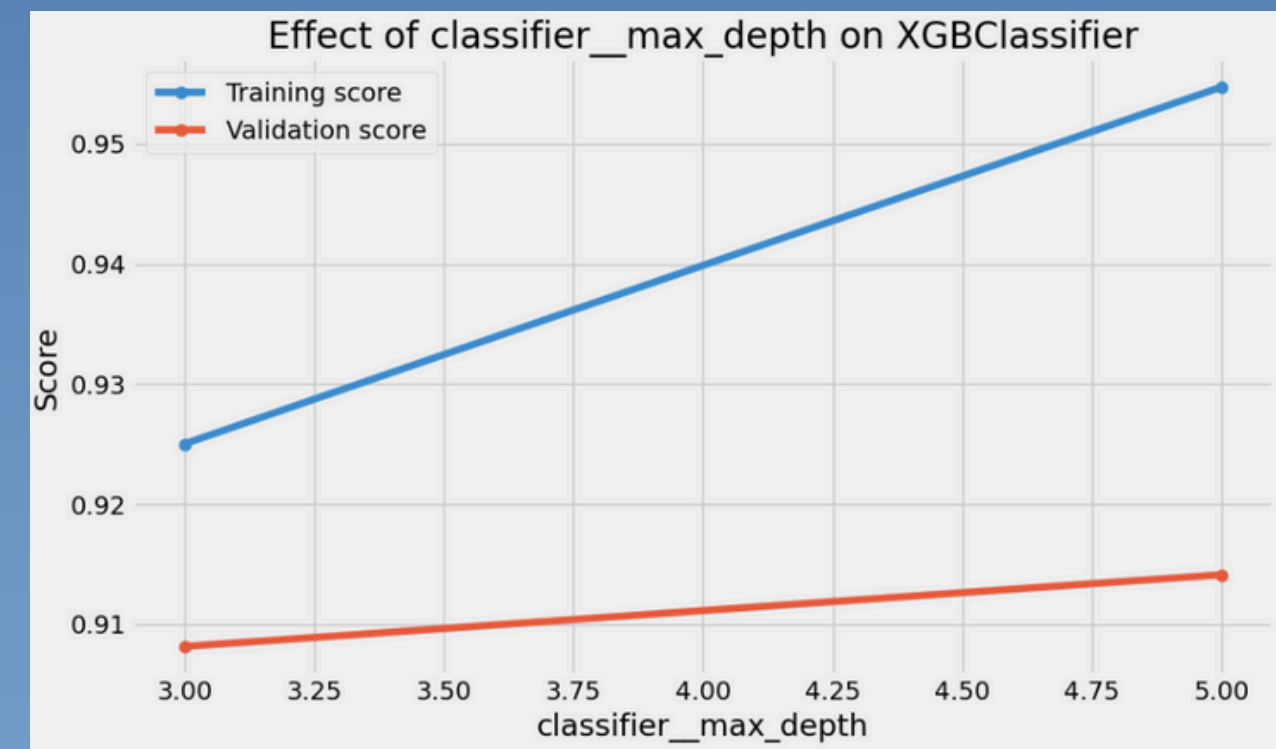
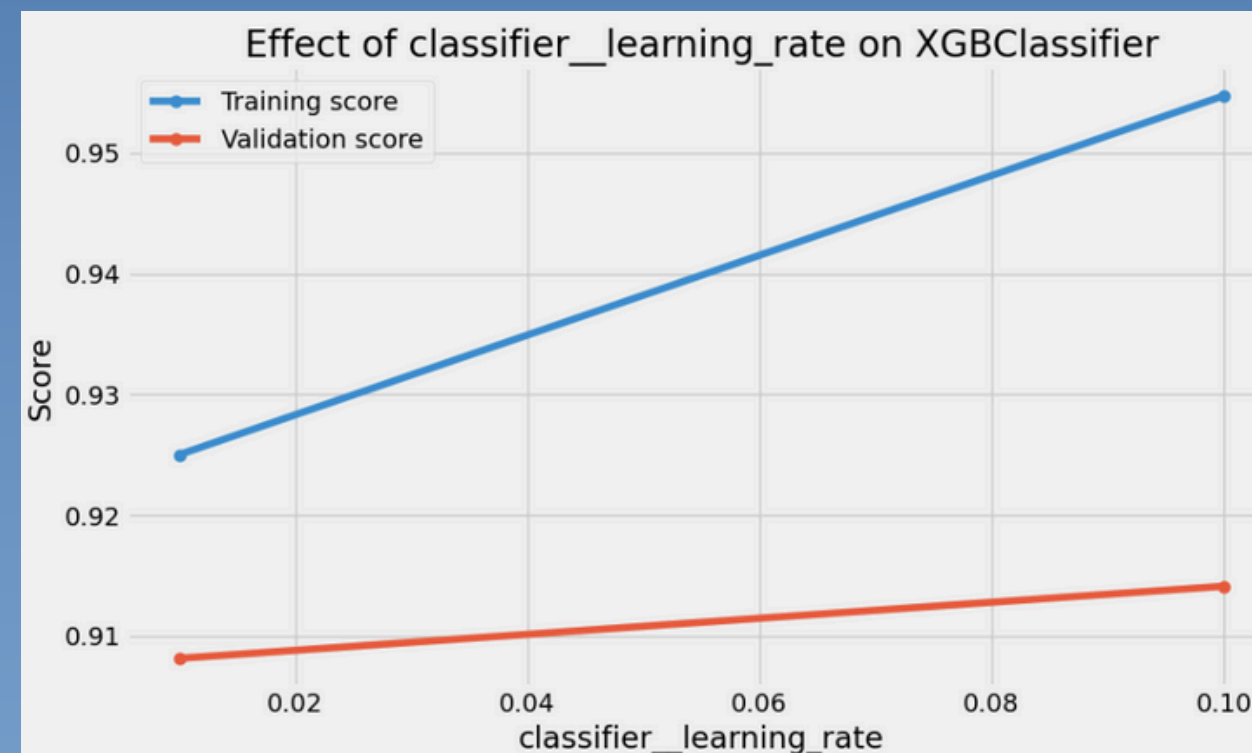
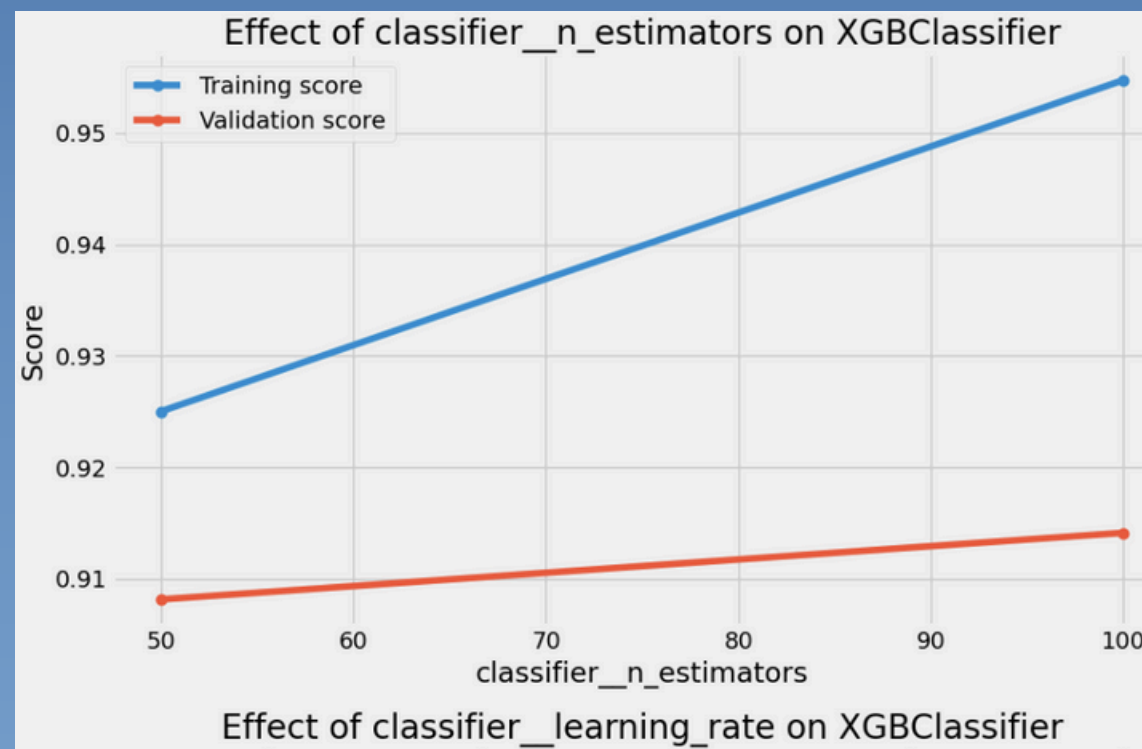
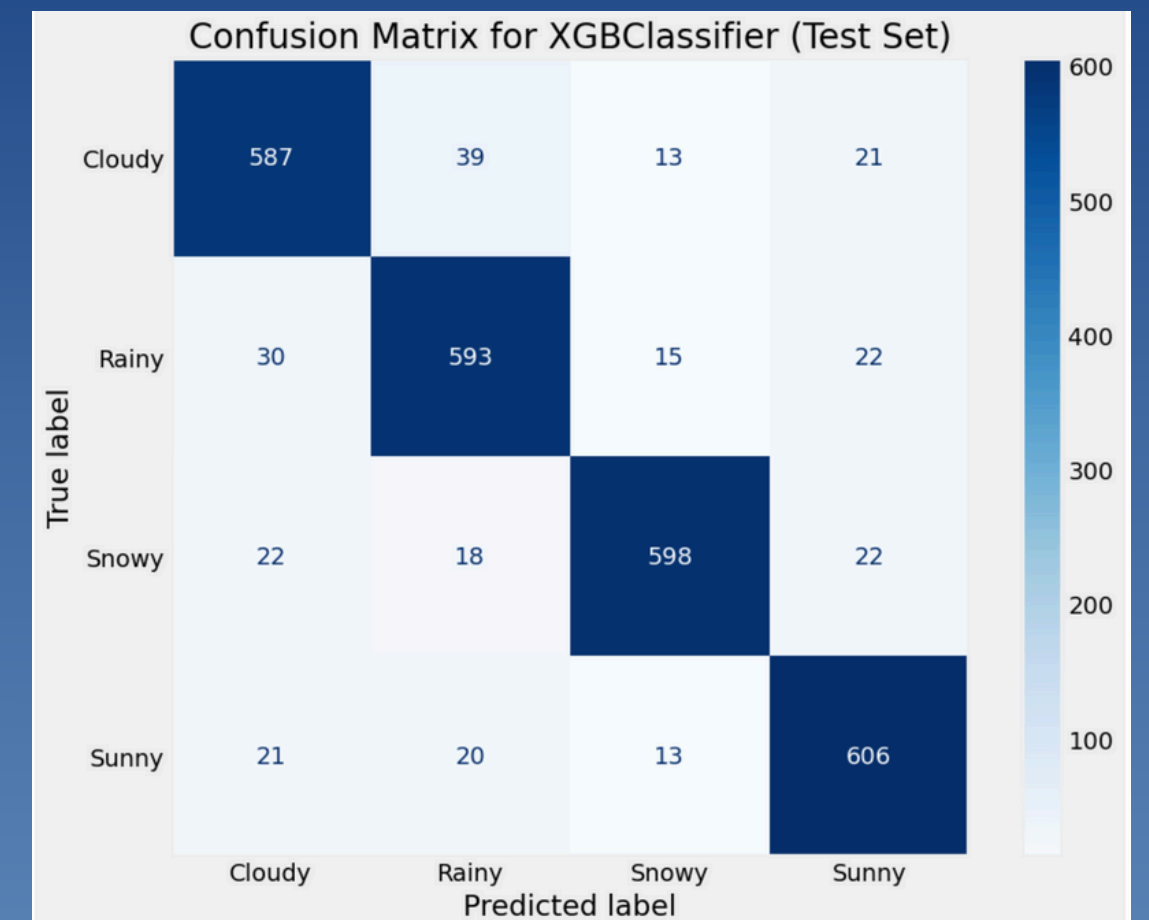
Performance Metrics:

- Accuracy: 90% (best overall performance so far).
- Precision: 0.88 - 0.94 (model correctly identifies positive cases).
- Recall: 0.88 - 0.93 (ability to capture all positive cases).
- F1-score: 0.88 - 0.92 (balance between precision and recall).

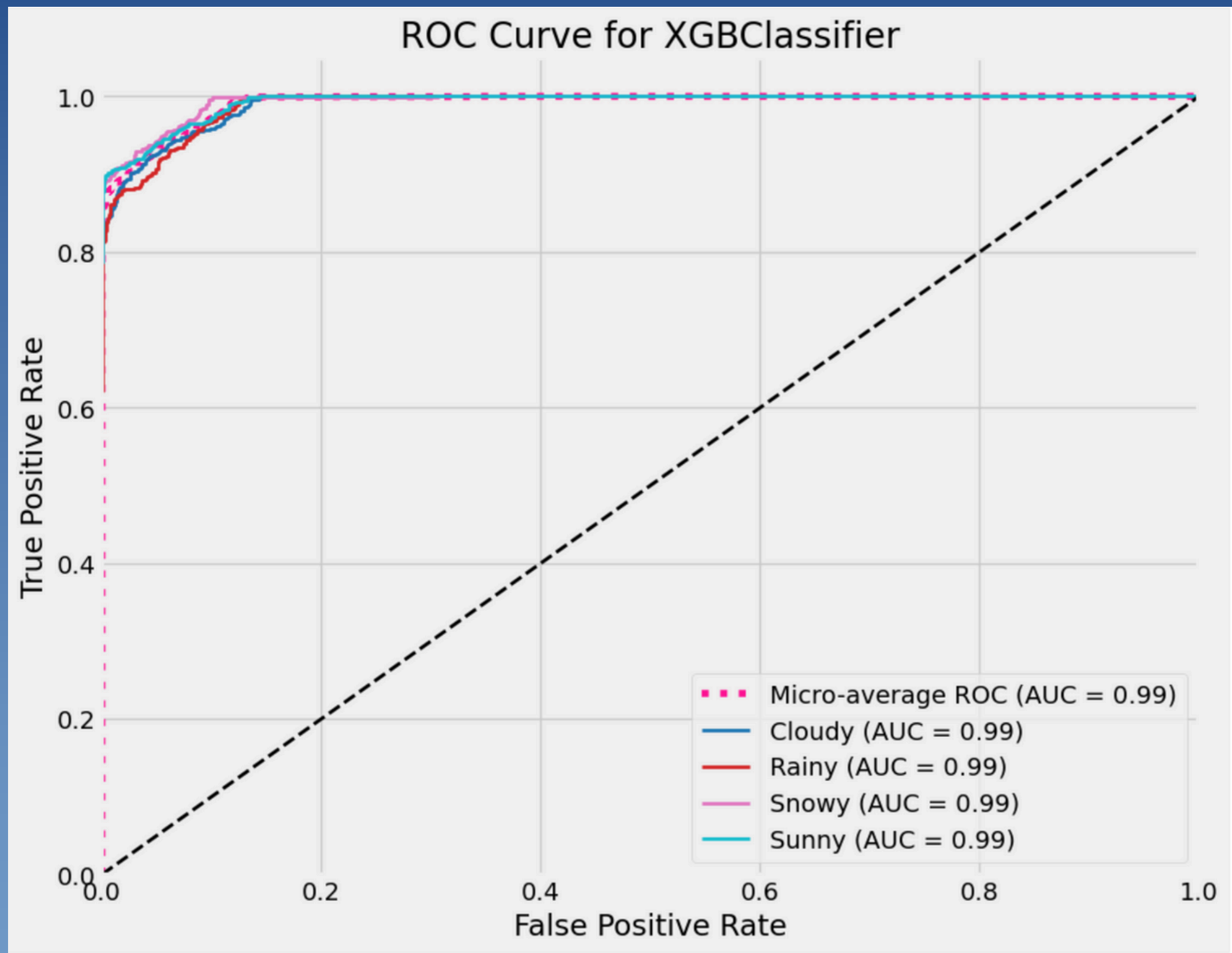
Gradient Boosting Classifier

Hyperparameters Used:

- Learning Rate: 0.1 (controls the step size during optimization).
- Number of Estimators: 50 (controls the number of boosting rounds).
- Max Depth: 5 (limits tree depth to prevent overfitting).



XGB Classifier



Classification Report for XGBClassifier (Test Set):				
	precision	recall	f1-score	support
Cloudy	0.89	0.89	0.89	660
Rainy	0.89	0.90	0.89	660
Snowy	0.94	0.91	0.92	660
Sunny	0.90	0.92	0.91	660
accuracy			0.90	2640
macro avg	0.90	0.90	0.90	2640
weighted avg	0.90	0.90	0.90	2640

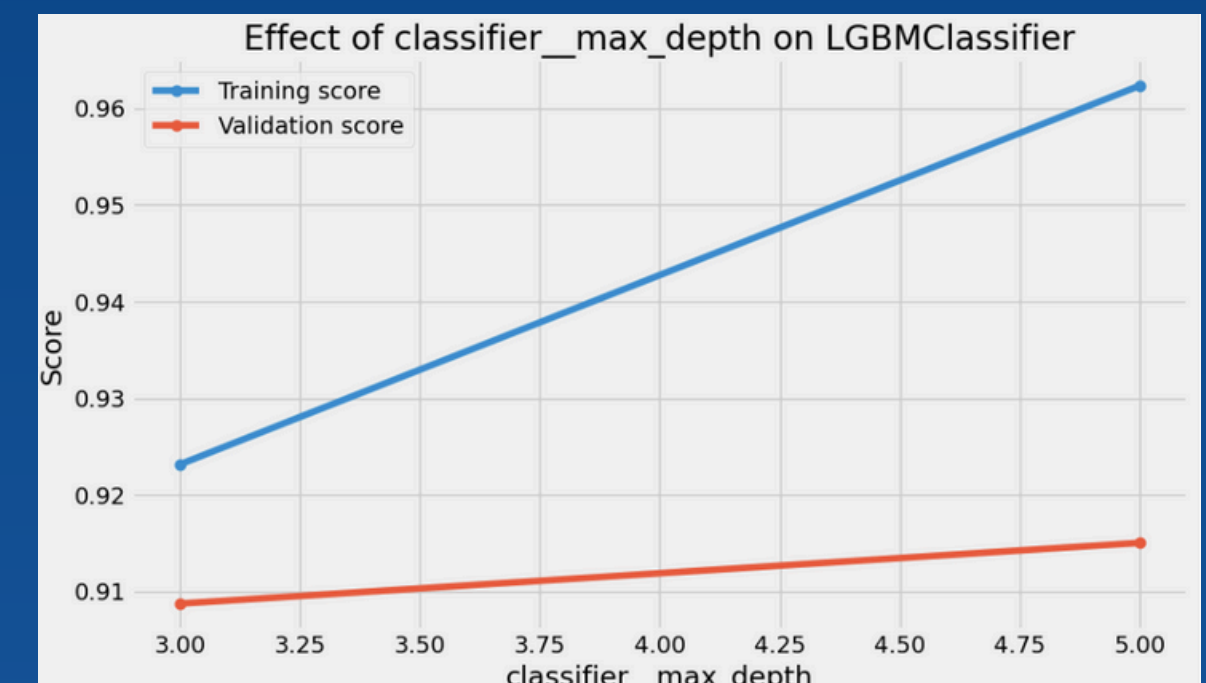
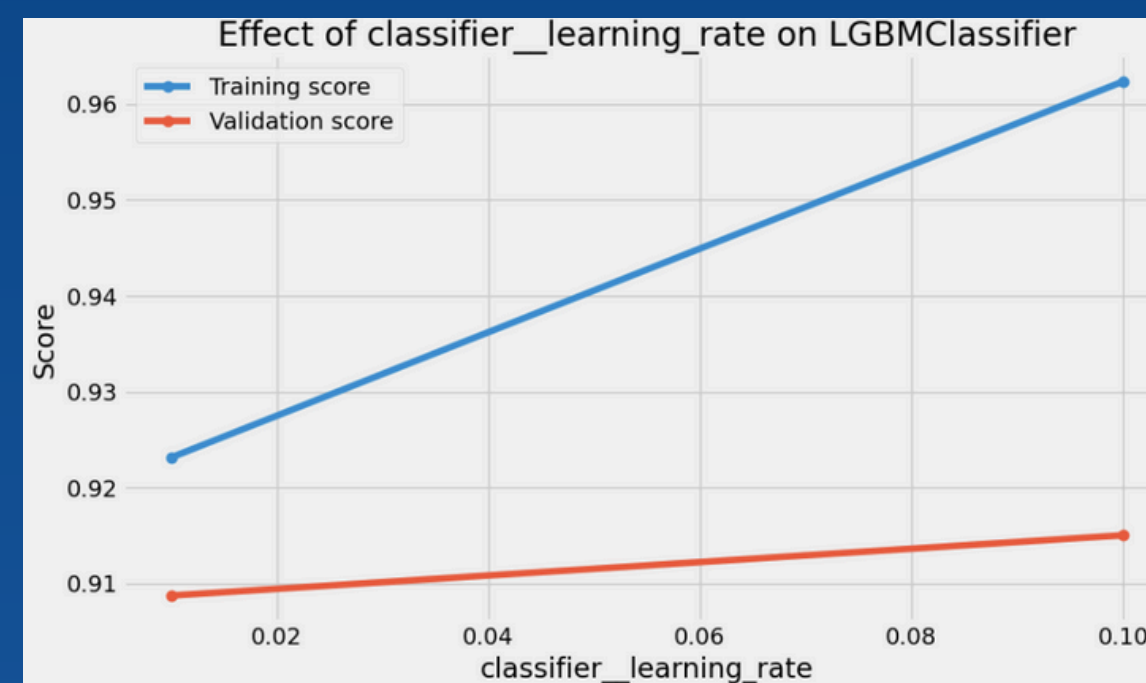
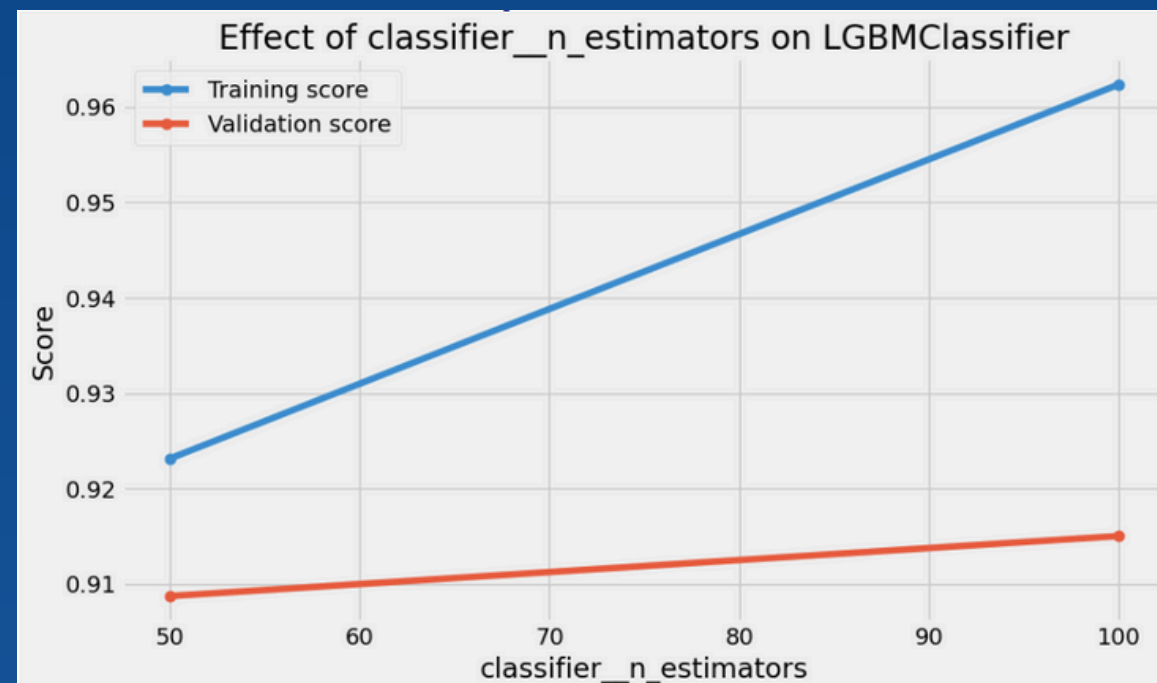
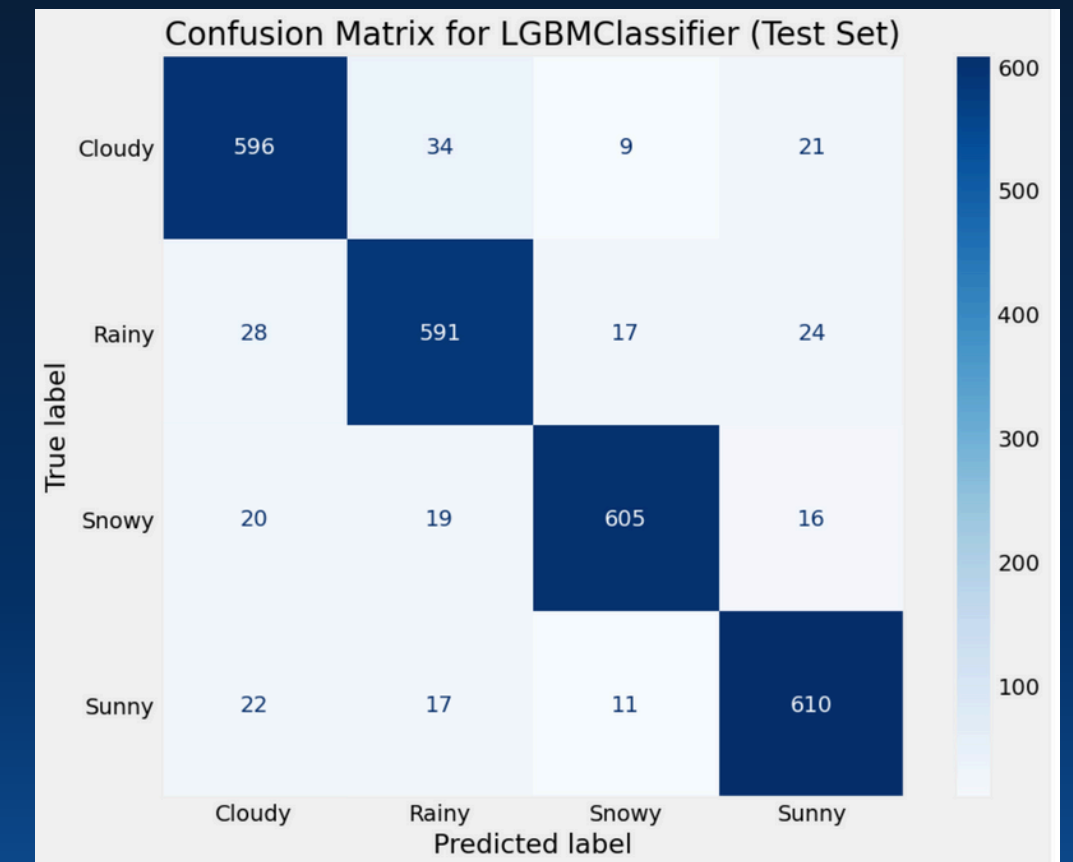
Performance Metrics:

- Accuracy: 90% (strong overall performance).
- Precision: 0.89 - 0.94 (model correctly identifies positive cases).
- Recall: 0.89 - 0.92 (ability to capture all positive cases).
- F1-score: 0.89 - 0.92 (balance between precision and recall).

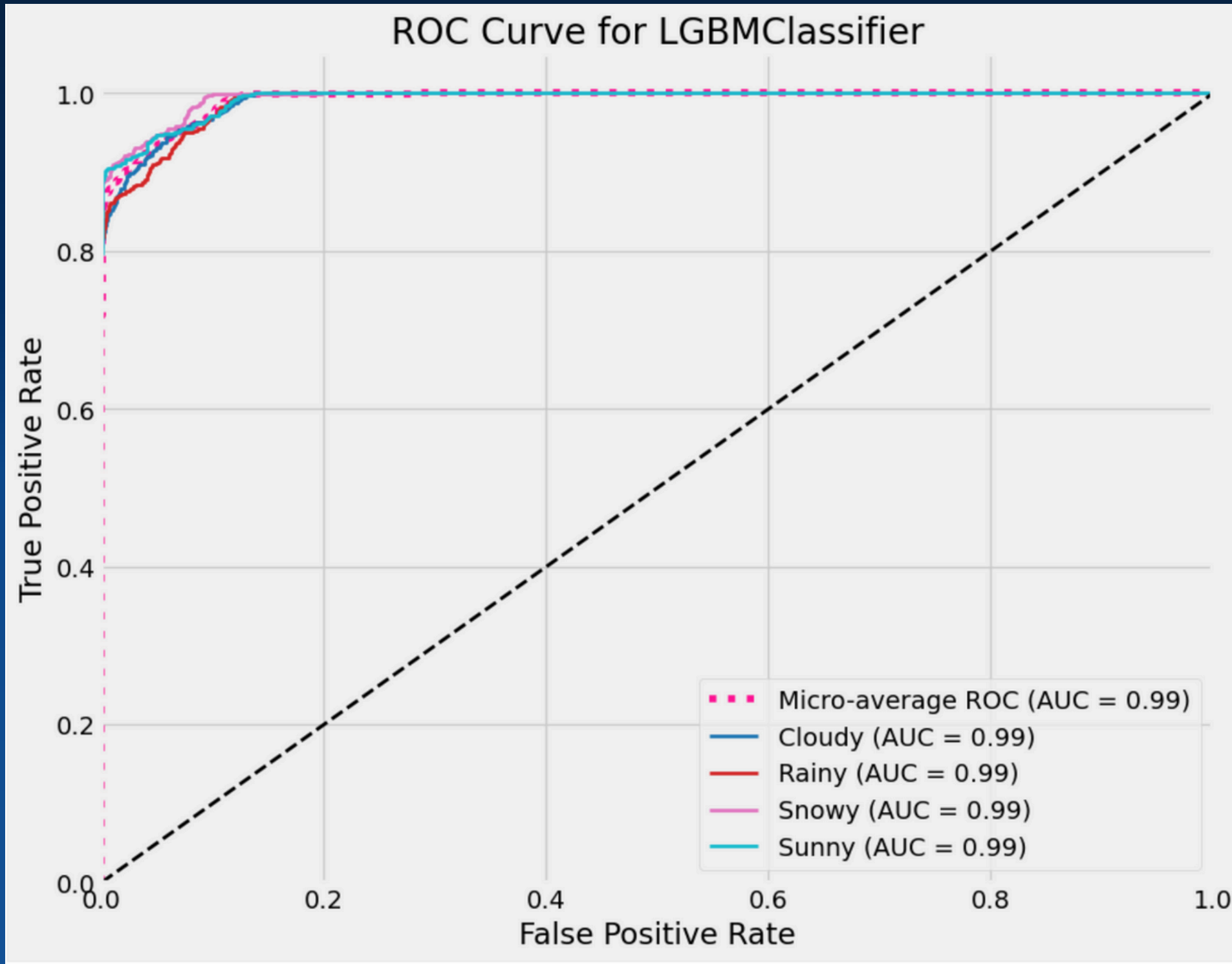
XGB Classifier

Hyperparameters Used:

- Learning Rate: 0.1 (controls step size during optimization).
- Number of Estimators: 50 (defines the number of boosting stages).
- Max Depth: 5 (limits tree depth to prevent overfitting).



LGBM Classifier



Classification Report for LGBMClassifier (Test Set):				
	precision	recall	f1-score	support
Cloudy	0.89	0.90	0.90	660
Rainy	0.89	0.90	0.89	660
Snowy	0.94	0.92	0.93	660
Sunny	0.91	0.92	0.92	660
accuracy			0.91	2640
macro avg	0.91	0.91	0.91	2640
weighted avg	0.91	0.91	0.91	2640

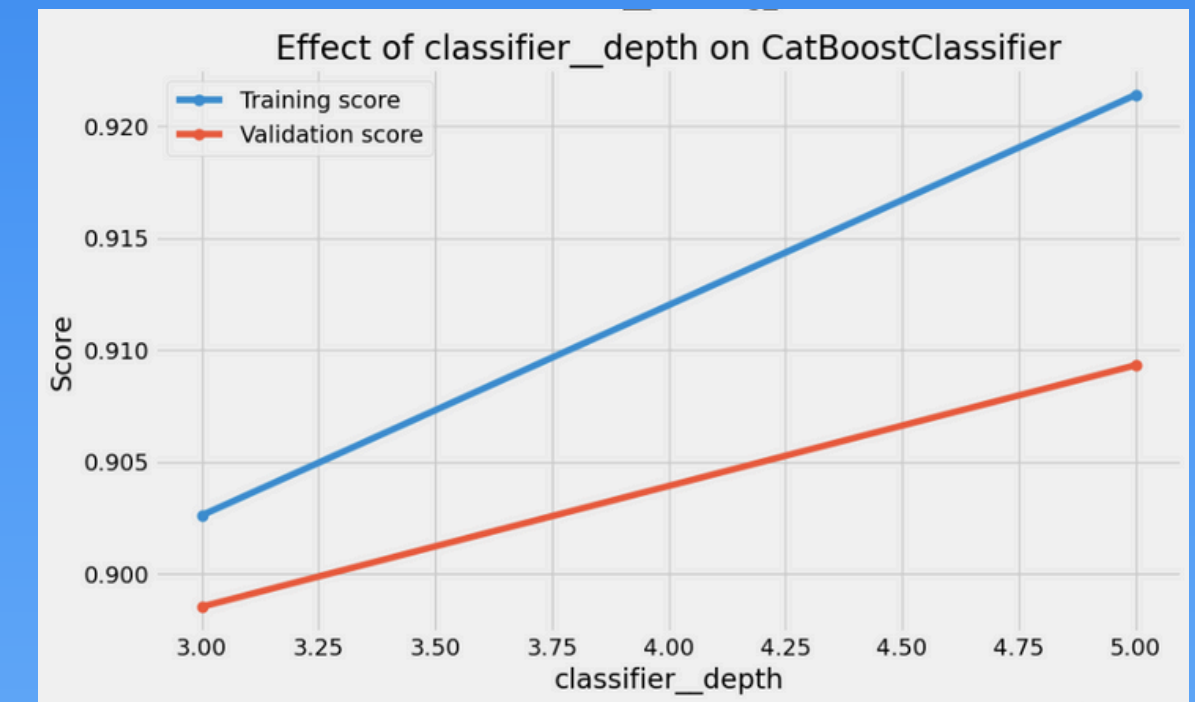
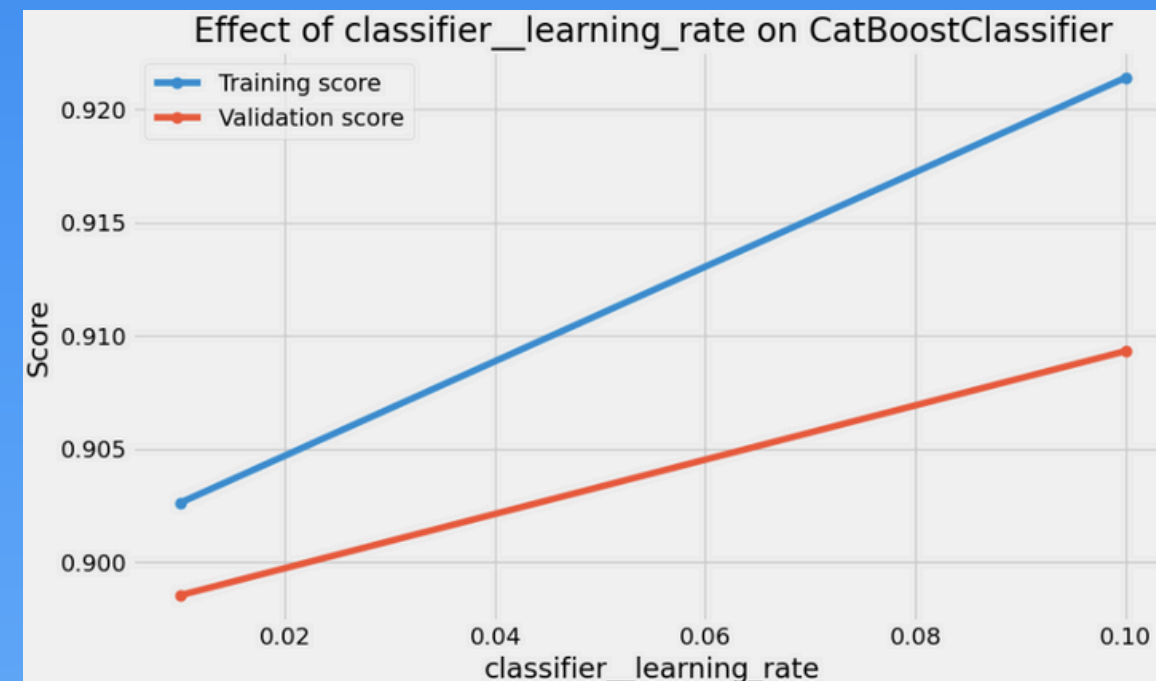
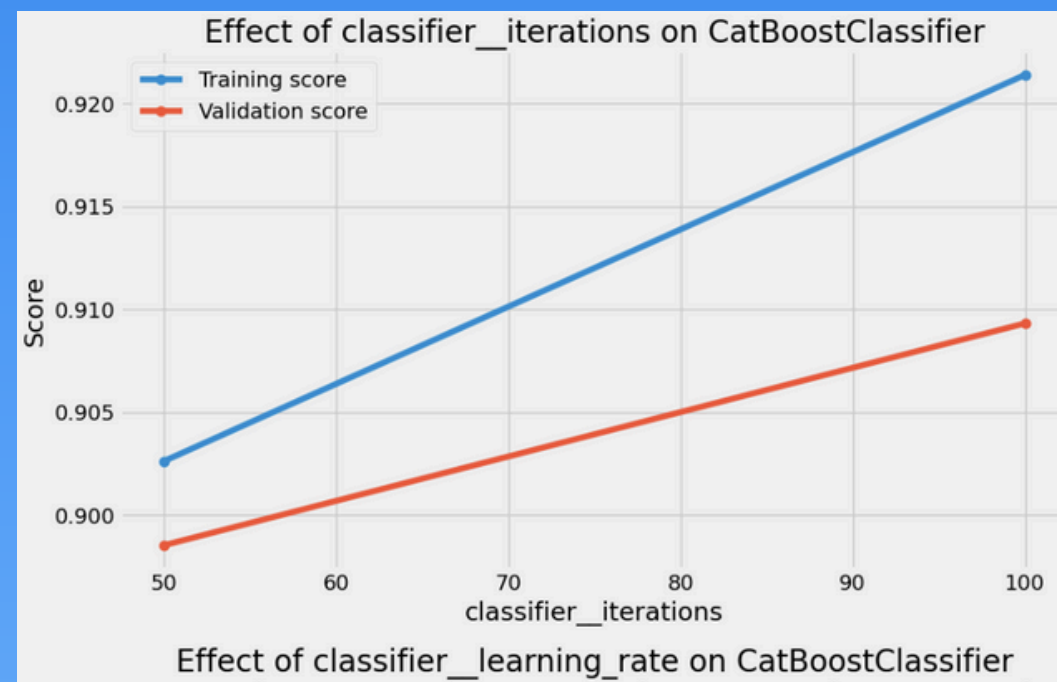
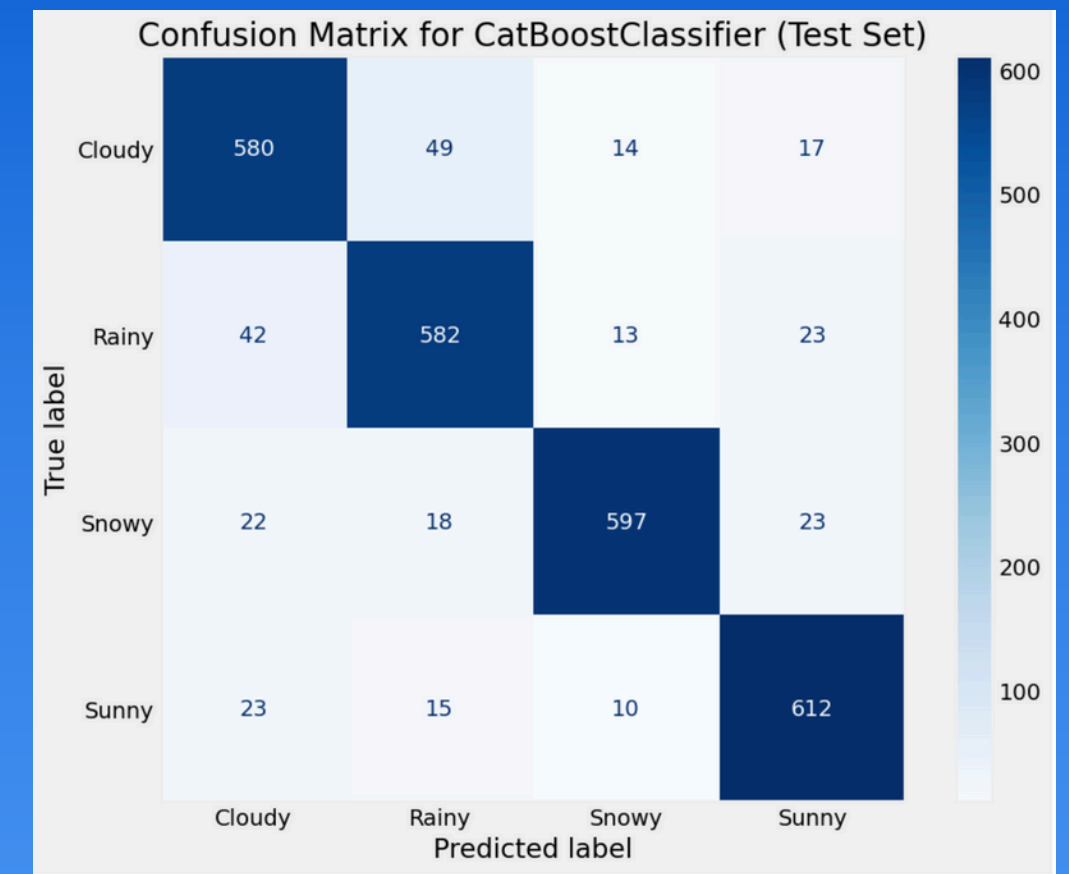
Performance Metrics:

- Accuracy: 91% (best overall performance so far).
- Precision: 0.89 - 0.94 (model correctly identifies positive cases).
- Recall: 0.90 - 0.92 (ability to capture all positive cases).
- F1-score: 0.89 - 0.93 (balance between precision and recall).

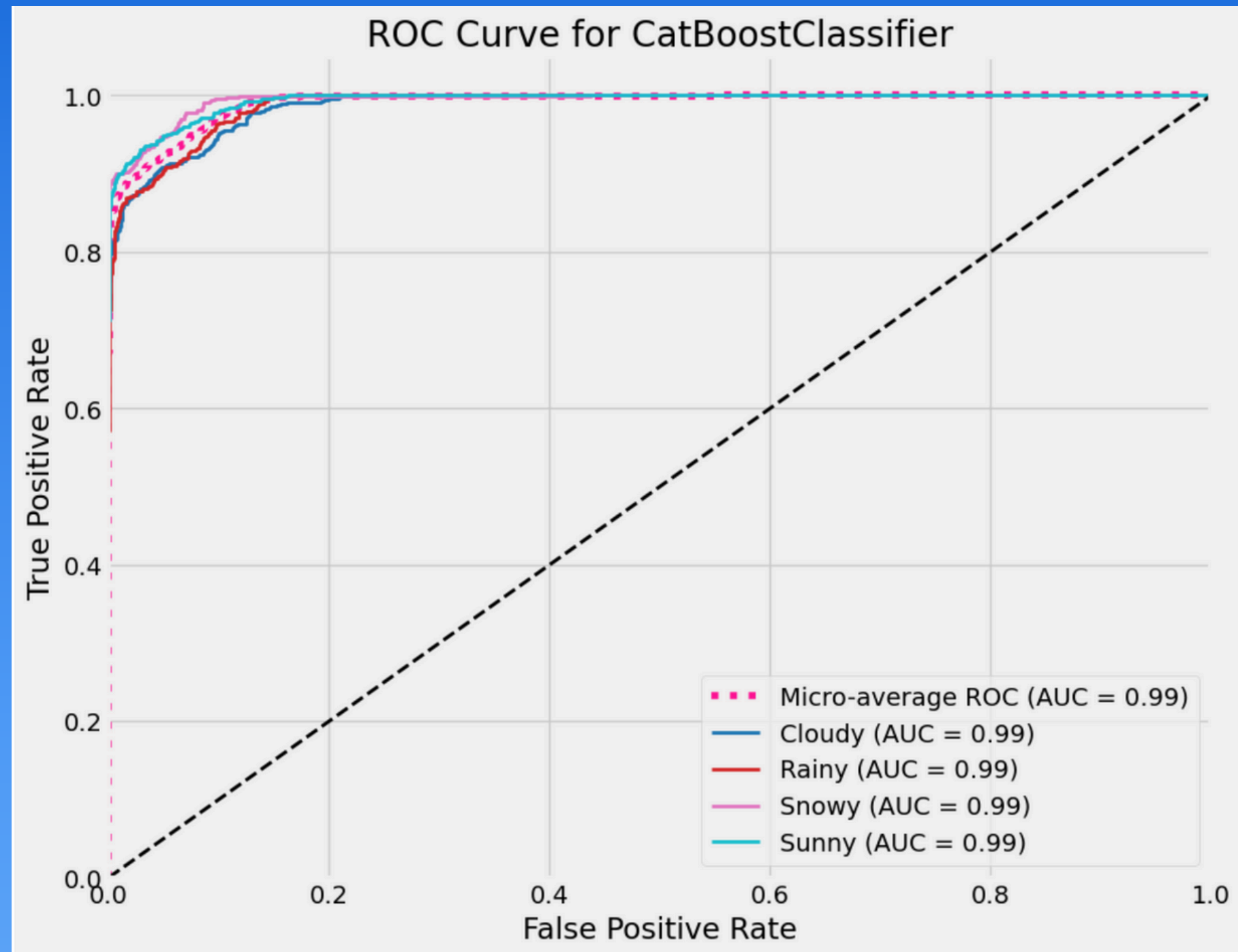
LGBM Classifier

Hyperparameters Used:

- Depth: 5 (controls the maximum depth of trees).
- Iterations: 100 (defines the number of boosting rounds).
- Learning Rate: 0.1 (adjusts step size for weight updates).



Cat Boost Classifier



Classification Report for CatBoostClassifier (Test Set):				
	precision	recall	f1-score	support
Cloudy	0.87	0.88	0.87	660
Rainy	0.88	0.88	0.88	660
Snowy	0.94	0.90	0.92	660
Sunny	0.91	0.93	0.92	660
accuracy			0.90	2640
macro avg	0.90	0.90	0.90	2640
weighted avg	0.90	0.90	0.90	2640

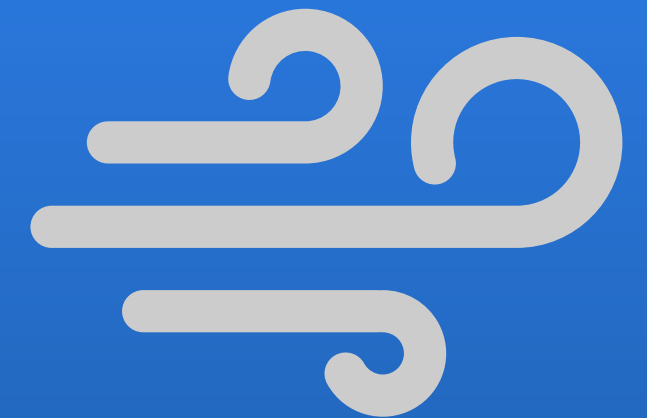
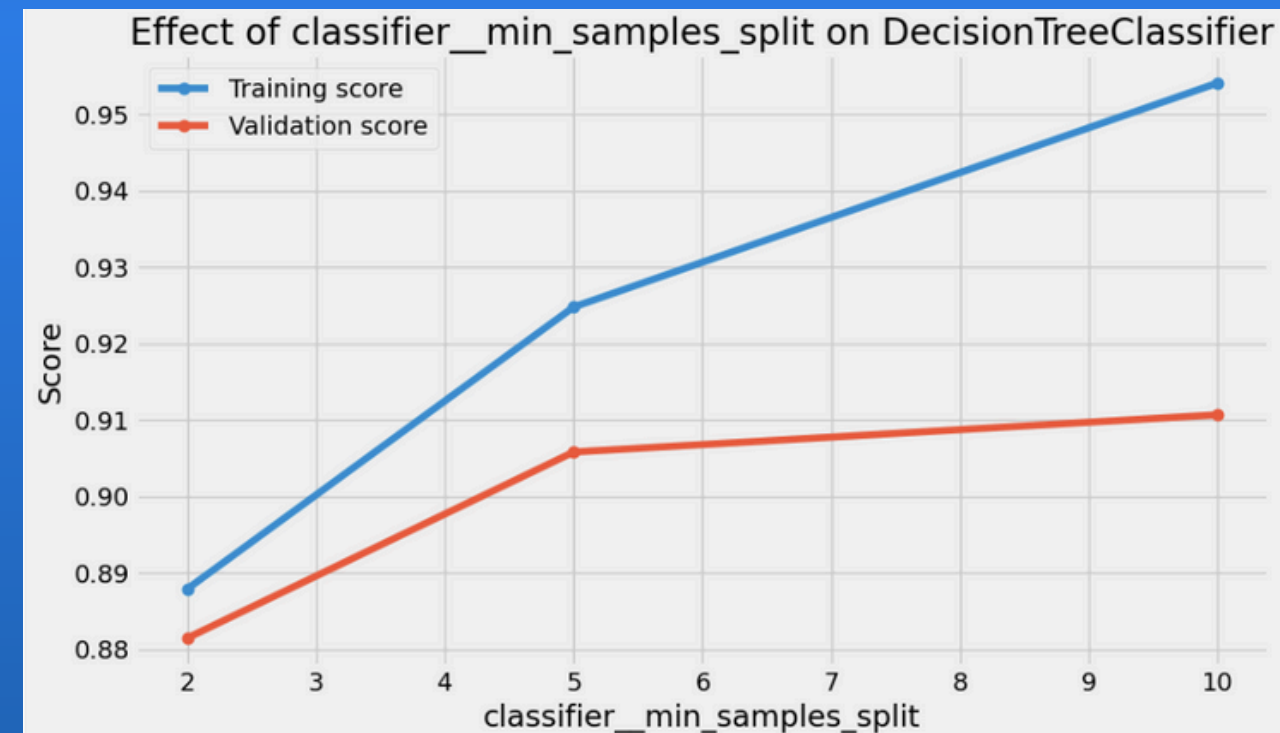
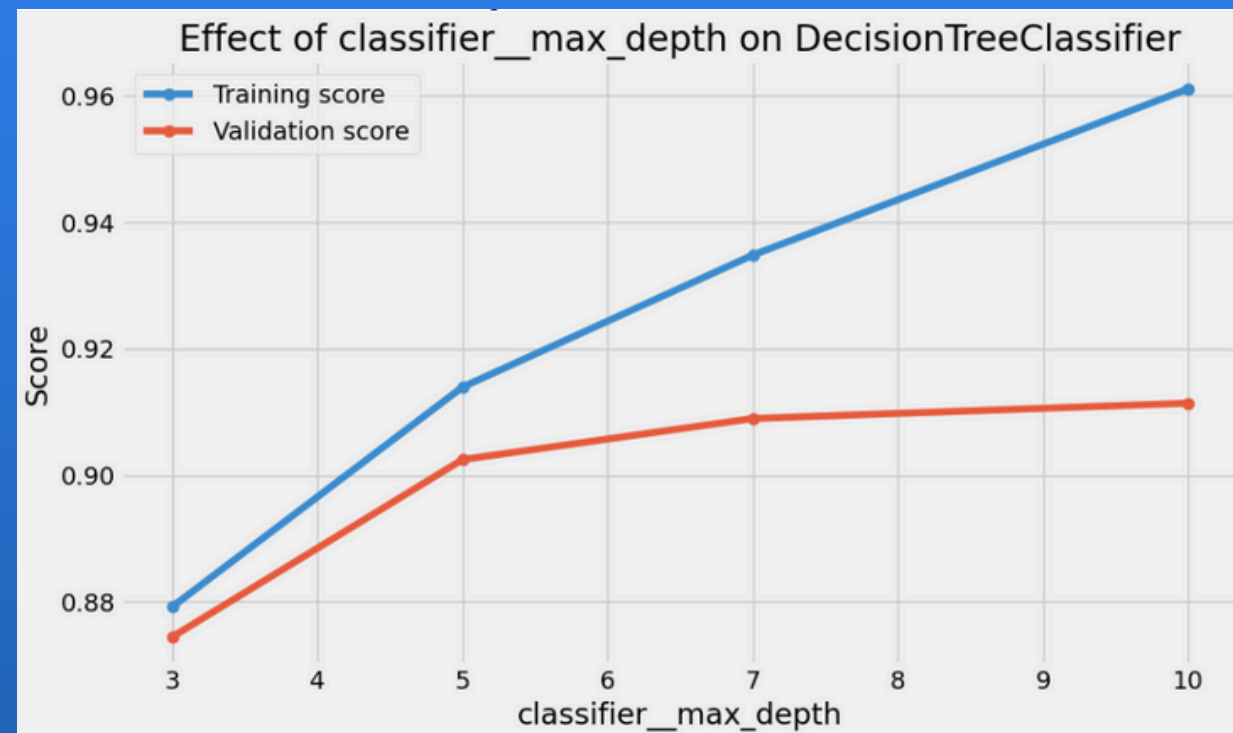
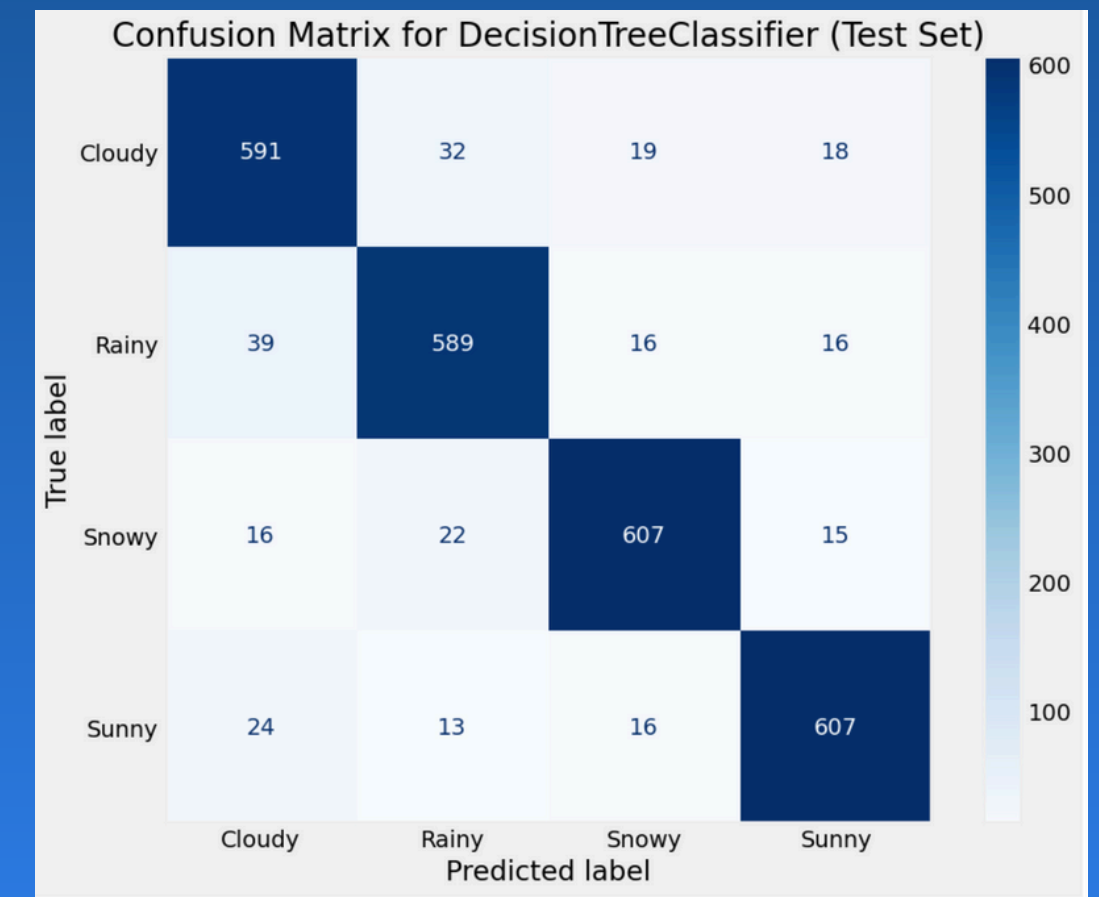
Performance Metrics:

- Accuracy: 90% (strong overall performance).
- Precision: 0.87 - 0.94 (model correctly identifies positive cases).
- Recall: 0.88 - 0.93 (ability to capture all positive cases).
- F1-score: 0.87 - 0.92 (balance between precision and recall).

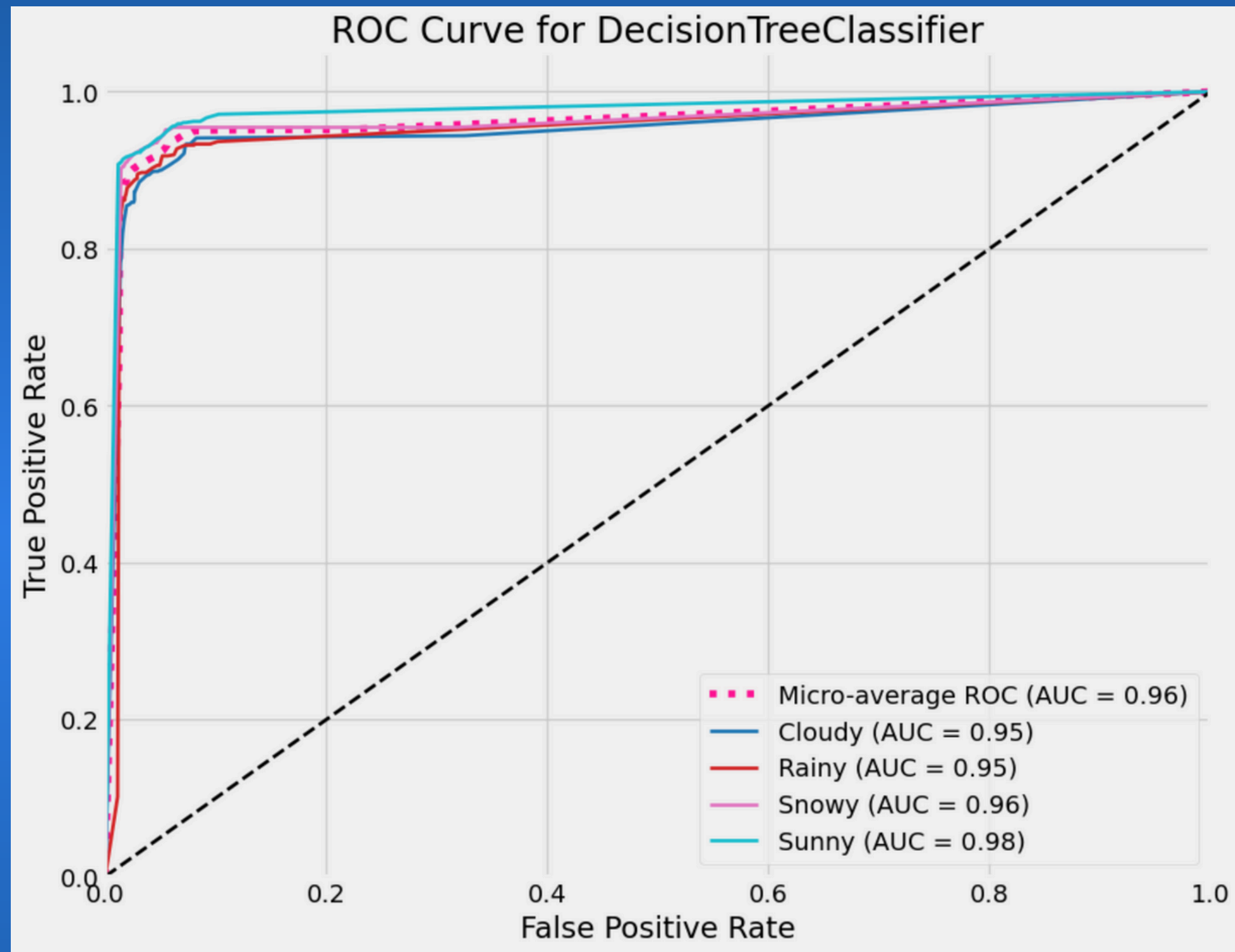
Cat Boost Classifier

Hyperparameters Used:

- Max Depth: 10 (limits tree depth to control overfitting).
- Min Samples Split: 2 (minimum samples required to split an internal node).



Decision Tree Classifier



Classification Report for DecisionTreeClassifier (Test Set):				
	precision	recall	f1-score	support
Cloudy	0.88	0.90	0.89	660
Rainy	0.90	0.89	0.90	660
Snowy	0.92	0.92	0.92	660
Sunny	0.93	0.92	0.92	660
accuracy			0.91	2640
macro avg	0.91	0.91	0.91	2640
weighted avg	0.91	0.91	0.91	2640

Performance Metrics:

- Accuracy: 91% (strong overall performance).
- Precision: 0.88 - 0.93 (model correctly identifies positive cases).
- Recall: 0.89 - 0.92 (ability to capture all positive cases).
- F1-score: 0.89 - 0.92 (balance between precision and recall).

Decision Tree Classifier

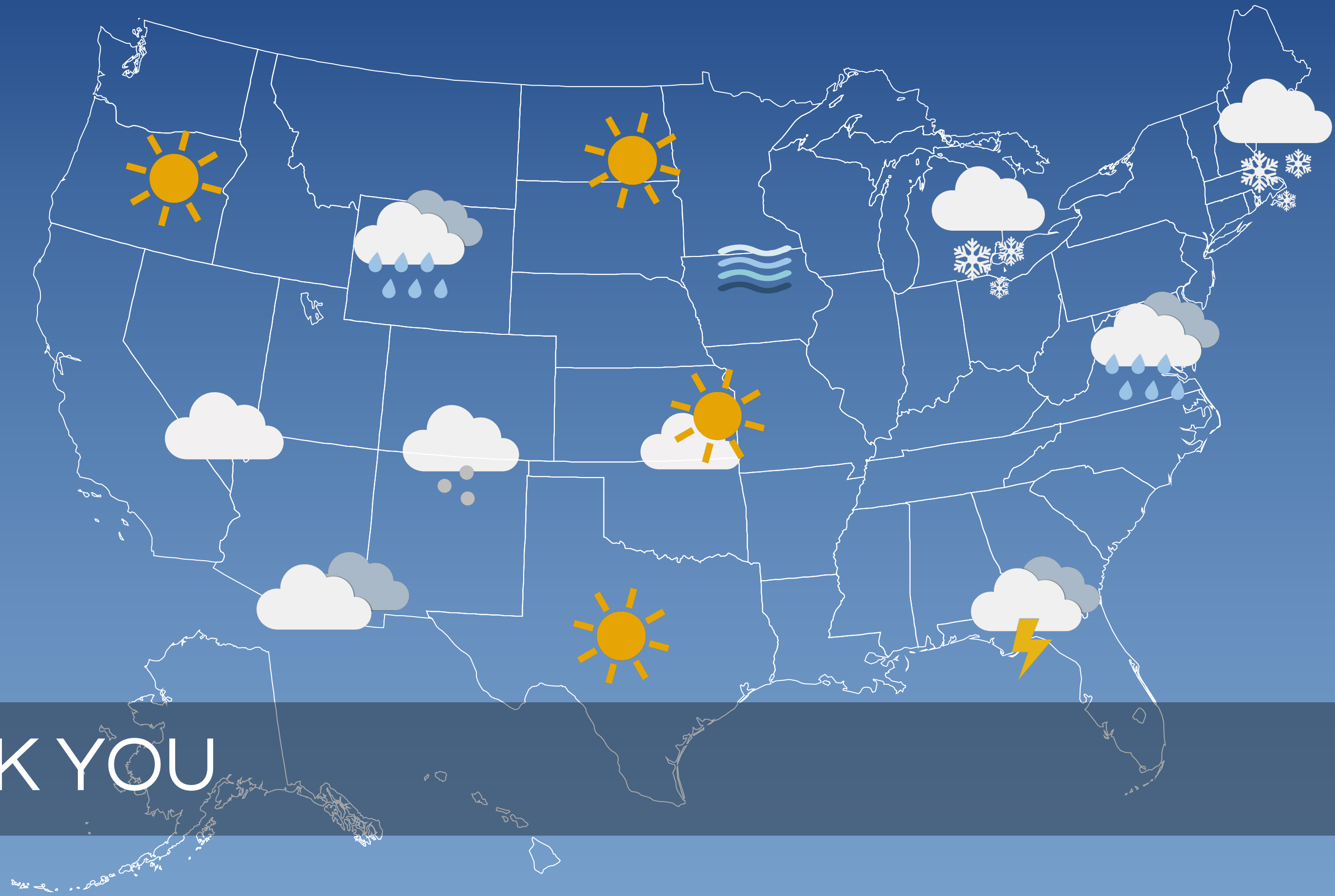


Model	Training Accuracy	Test Accuracy	Test F1 Score
LGBMClassifier	0.966961	0.909848	0.909916
DecisionTreeClassifier	0.965383	0.906818	0.906861
GradientBoostingClassifier	0.95181	0.904167	0.904257
XGBClassifier	0.964331	0.90303	0.903106
RandomForestClassifier	0.956229	0.898485	0.899283
CatBoostClassifier	0.940657	0.898106	0.89822
SVC	0.951073	0.894697	0.895129
KNeighborsClassifier	0.923822	0.874621	0.875087
LogisticRegression	0.875947	0.855682	0.855704

After evaluating all models, the **LGBM Classifier** model stands out as the best performer based on both testing accuracy and F1 score.

This indicates that the model not only predicts the weather types accurately but also maintains a good balance between precision and recall, making it a reliable choice for this classification task.

Final Model Evaluation & Selection



THANK YOU